

A robust home health care routing-scheduling problem with temporal dependencies under uncertainty

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ABSTRACT

Inspiring by real-world assumptions, this paper aims to present a robust optimization model for a home health care routing-scheduling problem with uncertain service and travel times. The constraints include temporal precedence and synchronization constraints, timing constraints for the transferring biological samples, and constraints associated with the multiple deployments of caregivers to provide the possibility of visiting one patient multiple times in the one-day planning horizon. Also, the existing uncertainty in service and travel times arising from an increase in customer-oriented service strategies is another crucial issue that should be appropriately addressed. Due to the complexity of this problem, three meta-heuristic algorithms (i.e., simulated annealing, genetic algorithm, and memetic algorithm) are proposed to solve the developed model. A series of experiments are implemented and shown that the memetic algorithm outperforms other proposed algorithms for large-sized problems. Furthermore, a detailed analysis of the results achieved by solving deterministic and robust models demonstrates the advantage of using the robust model. Finally, the dynamic version of this problem is proposed, and two dispatching policies are used to address this problem.

1. Introduction

According to the World Health Organization, the rate of elderly people that need different services of care has been increased dramatically (Mankowska, Meisel, & Bierwirth, 2014). In 1958, this social issue led to the foundation of Danish companies, whose primary purpose was to provide different home services for disabled patients, who cannot stay in hospitals for a long period to recover them from injuries and illnesses (Rasmussen, Justesen, Dohn, & Larsen, 2012). Different advantages of home health care (HHC) are considered as a treatment in one's home, fewer hospitalization and lower demand for hospital admissions, oriented health care systems to provide their services through HHC companies rather than hospitals (Cissé, et al., 2017). In 2011, around 143,600 persons were employed and paid through 12,200 HHC companies to provide services for more than 4.7 million patients in the United States (Fikar and Hirsch, 2017). Also, it is estimated that up to 5% of the total public budget in the health sector is spent on HHC services (Genet, Boerma, Kroneman, Hutchinson, & Saltman, 2012). One of

the most important sectors of HHC companies is logistics activities, which aim to deliver drugs or medical services from a hospital to patients' homes, take biological samples, and transfer them to the laboratory through qualified nurses within the given time windows (Liu, Xie, Augusto, & Rodriguez, 2013). According to a survey conducted by two municipalities in Norway, on average 22% of the nurses' working time is spent on driving time (Holm & Angelsen, 2014). Therefore, routing and scheduling problems arising from these social and economic aspects of HHC need to be addressed efficiently and practically (Moussavi, Mahdjoub, & Grunder, 2019).

A home health care routing-scheduling problem (HHCRSP) differs from a classic vehicle routing problem with simultaneous delivery and pickup and time windows (VRPSDPTW) in several following aspects. First, according to the patients' physical condition and their demand diversity for other services, the services delivered by HHC companies can be divided into two categories: The first and more important one is medical services that patients can request at different hours of a day based on their doctor's prescriptions. For instance, according to the

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American Diabetes Association, patients diagnosed with diabetes may need to have an insulin injection up to four times a day, and it should be administered within certain time windows (Fikar & Hirsch, 2017). The second type of service is related to regular works done in housekeeping, such as assistance for cleaning, tidying, preparing food, and aids for other tasks. Due to this demand diversity, timing constraints in our study are not only limited to time windows but also interconnection between the routes of caregivers needs to be addressed as time constraints. This interconnection is classified as operations synchronization (Drexler, 2012).

To be more specific, there are two kinds of synchronization (Labadie, Prins, & Yang, 2014): simultaneous and precedence synchronization. Simultaneous synchronization of two visits is applicable when a patient requires a bath or getting in or out of bed. In this situation, two caregivers should perform the required tasks simultaneously. Patients also based on their health condition can ask for more ordered tasks that some services should be performed before other ones. This kind of visit is known as precedence synchronization. Precedence services usually occur in medical services that need to be performed under a full continuity of care. Continuity of care guarantees that each patient is allocated to a specified set of caregivers. Similarly, full continuity of care occurs when a unique caregiver is assigned to one patient over the planning horizon (Cissé et al., 2017).

Therefore, one of the most important characteristics of the HHCRSP, which has been neglected in almost all previous works, is the dependency between precedence synchronized services and the appropriate level of continuity of care. This means when a caregiver is assigned to one patient for performing a specific medical service, following precedence services related to the first service should be operated by the same caregiver in a one-day planning horizon. Keeping the same caregiver can reduce the losses of patients' medical information and more importantly, raise the level of confidence between the patient and caregiver (Freeman & Hughes, 2010). In another contrast with the classic VRPSDPTW, there is perishability for biological samples from some patients whose care procedure needs to be tracked. For example, for glucose measurements, the allowable time for centrifuging the drawn blood samples is 20–35 min (Tolonen, Kuulasmaa, Laatikainen, & Wolf, 2002). Hence, one caregiver has limited time to deliver biological samples to the laboratory. Another issue that needs to be tackled is limited human resources in the HHCRSP. To deal with this problem, proposing a mathematical model that can construct different routes and plans for staff members in different periods of a day makes the problem more practical and realistic. The need to simultaneously consider the collection of biological samples and the multiple deployments of nurses becomes apparent when, in the COVID-19 pandemic, PCR tests need to be performed in patients' homes at different times of the day. Inspired by these real-world assumptions, our planning problem is the first research that considers a collection of biological samples, their perishability, and multiple deployments of nurses simultaneously.

There are other critical elements in HHC services, including service and travel times. Common factors (e.g., varying diagnosing time, park places, rush hours and, weather) lead to the uncertainty of the service and travel times (Shi, Boudouh, & Grunder, 2019). To deal with uncertain parameters, stochastic approaches are the most well-known methods (Boronoos, Mousazadeh, & Torabi, 2020). These methods require uncertain parameters with the known probability distributions and have been applied to a wide range of vehicle routing problem (VRP) variants. However, these approaches remain useless when the probability of uncertain parameters cannot be determined precisely (Amini, Tavakkoli-Moghaddam, & Ebrahimnejad, 2020). In contrast with the stochastic method, without distinguished probability distributions of uncertain parameters, robust optimization can find efficient solutions for all possible scenarios (Rouky, Boukachour, Boudebous, & Alaoui, 2018). Therefore, due to the lack of enough data to estimate these distributions, we attempt to propose a robust strategy for HHC company's

services using a robust optimization model. To the best of our knowledge, this is the first study that proposes a robust optimization model with uncertain travel and service times for the HHCRSP by considering the mentioned real-world constraints related to this context.

The HHCRSP can also be categorized as static or dynamic (Demirbilek, Branke, & Strauss, 2019). When the input data is provided in advance and does not depend on time, the problem is considered to be static. In contrast, there is some time-dependent information in the dynamic version of the problem (e.g., a new request may appear) while the system is already running, and the routing procedure needs to be performed again to support the new request properly. In this paper, postponement and greedy policies are used to address the dynamic version of the problem that allows new requests to be handled.

Since the proposed problem is entirely complicated, we develop a new algorithm that can find an effective initial solution according to constraints in a very short period. Then, we combine this algorithm with simulated annealing (SA), genetic algorithm (GA), and memetic algorithm (MA) for finding near-optimal solutions for different deterministic instances taken from the Solomon (1987) benchmark. Then, MA is used to solve the robust model, and the advantage of using this model is highlighted using the Monte Carlo simulation. Finally, postponement and greedy policies are used to address the dynamic version of the proposed HHCRSP.

This study contributes to the HHCRSP with the following aspects.

- (1) Considering the dependency between precedence synchronized services and appropriate level of continuity of care.
- (2) Considering multiple deployments of one HHC caregiver and the possibility of visiting one patient multiple times during different periods of a day.
- (3) Presenting a robust optimization model for the HHCRSP with the mentioned features and uncertain travel and service times.
- (4) Developing three efficient *meta*-heuristic algorithms (i.e., SA, GA, and MA) for solving the developed model.
- (5) Developing the dynamic version of the proposed problem using postponement and greedy dispatching policies.

The remainder of this study is organized as follows. Section 2 reviews the relevant literature. Section 3 presents the developed model with the problem description. Section 4 describes the detailed steps of the proposed *meta*-heuristic algorithms. Section 5 illustrates the extensive computational experiments with the evaluation of the proposed model and algorithms. Finally, Section 6 states the conclusion and some directions for future studies.

2. Literature review

In general, the related literature can be categorized based on the planning horizon. The first category is related to problems with a one-day planning horizon, and the second one is attributed to problems with more than the one-day planning horizon.

Based on this classification, we first review papers dealing with a short planning horizon. Bredström & Rönnqvist (2008), Kergosien, Lenté, & Billaut (2009), Rasmussen et al. (2012), Mankowska et al. (2014), Haddadene, Labadie, & Prodhon (2016), and Cappanera, Requejo, and Scutellà (2020) considered both simultaneous and precedence synchronization assumptions in their models. Bredström & Rönnqvist (2008) and Rasmussen et al. (2012) applied the duplication procedure for customer nodes to address synchronization constraints. Rasmussen et al. (2012) implemented a dynamic column generation-based algorithm (i.e., branch-and-price) to solve a set partitioning model derived from the decomposition of a home care crew scheduling problem for a one-day planning horizon. In contrast, Haddadene, Labadie, & Prodhon (2016) developed a mathematical model to deal with the increased size of the problem caused by duplicating customer nodes. They also hybridized the instruction of the greedy randomized adaptive

search procedure (GRASP) and the iterative local search (ILS) algorithms for large-sized instances. Although Rasmussen et al. (2012) and Haddadene, Labadie, & Prodhon (2016) could manage to model both synchronization constraints in one model, Bredström & Rönnqvist (2008) considered two separated constraints to deal with them. Manowska et al. (2014) applied an adaptive variable neighborhood search (VNS) for the daily scheduling of home care services. The proposed objective function optimizes both economic and service-oriented aspects of the problem by minimizing the total traveling cost and tardiness of services. Cappanera, et al. (2020) proposed some lower bounding techniques with effective valid inequalities to solve their generalized Skill VRP mixed-integer linear programming model.

Some papers have considered synchronization constraints as a general concept and dissimilar to reviewed papers, and they have not been categorized into different types. Eveborn, Flisberg, & Rönnqvist (2006) introduced Laps Care as a decision support system to provide efficient visiting schedules for care staff in a Sweden HHC system. Similar to Rasmussen et al. (2012), the problem was formulated as a set partitioning model and solved by a repeated matching algorithm. They addressed simultaneous synchronization constraints by splitting visits, which require more than one caregiver. Decerle, Grunder, El Hassani, & Barakat (2018) brought more flexibility to the HHCRSP by proposing soft constraints for time windows and synchronization. The duplication procedure for addressing synchronized services was implemented by assigning one node for each visit. For instance, if one patient asks for synchronized services, two nodes are created for this patient with the same location information. The validation of the proposed MA was evaluated by the existing benchmark on the instances of Bredström & Rönnqvist (2008). In further research, Decerle, Grunder, El Hassani, & Barakat (2019b) applied a Pareto-based optimization algorithm to analyze a trade-off between the three conflicting objectives: minimization of the total working time of the care providers, maximization of the service level, and minimization of the maximal working time difference for staff members. Decerle, Grunder, El Hassani, & Barakat (2019a) proposed a hybrid memetic-ant colony optimization algorithm for the HHCRSP. They considered different features for their problem, including synchronization constraints, time windows, and caregiver skills.

Among papers that considered a one-day planning horizon, some authors have focused on a new variety of the HHCRSP to tackle the problem of collecting some products, such as biological samples and unused medical devices, and then delivering them to hospitals and laboratories. Liu et al. (2013) introduced a different variant of the VRPSDPTW with the following demands: delivery from a hospital or depot to a patient independently and pick up from a patient to a medical lab or depot. The proposed GA and tabu search (TS) algorithms for solving two mixed-integer programming (MIP) models were tested on VRPTW test instances taken from Solomon (1987) and Gehring & Homberger (1999). Shi, Boudouh, & Grunder (2017) considered demand as a non-deterministic variable. A proposed fuzzy chance-constraint model was solved by a hybrid GA integrated with the stochastic simulation method. A simulation procedure was used to determine the best value of the dispatcher preference index (DPI) parameter. In more recent research Shi, Boudouh, Grunder, & Wang (2018) added uncertainty to travel and service times. Initially, five solution methods (Gurobi solver, SA, firefly algorithm, bat algorithm, and hybrid GA) have been applied to a deterministic model to find the most efficient algorithm. Subsequently, a proposed stochastic programming model was solved by the SA-based heuristic algorithm. Fathollahi-Fard, Hajiaghahi-Keshteli, & Tavakkoli-Moghaddam (2018) addressed green aspects of the HHCRSP by proposing a bi-objective optimization model, in which a trade-off between the transportation costs and environmental pollution caused by vehicles' CO₂ emissions has been examined. In another study, Fathollahi-Fard, Hajiaghahi-Keshteli, & Tavakkoli-Moghaddam (2018) presented a lagrangian relaxation-based method to solve the developed HHCRSP mathematical model. Shahnejat-

Bushehri, Tavakkoli-Moghaddam, Momen, Ghasemkhani, and Tavakkoli-Moghaddam (2019) recently applied SA and TS to solve their proposed HHCRSP. Their suggested planning takes into account limited allowable times for transporting biological samples, and they also consider different precedence and synchronization constraints.

Subsequently, in the second category, we review papers dealing with a long planning horizon. Duque, Castro, Sörensen, & Goos (2015) proposed a planning problem that each patient can be allocated several visits per week. Each patient can be visited up to five times with the following frequency constraint. If the number of required visits is less than five, the patient should not be visited on consecutive days. They considered four different time slots that patients can be visited on their preferred time slots for weekly visits. Similarly, this preference characteristic has been considered for caregivers by being scheduled to work in their preferred time horizons. Two proposed objective functions (i.e., maximizing the preference and satisfaction factors and minimizing the total traveled distance) were considered. A lexicographic approach was proposed for a solution procedure. Bowers, Cheyne, Mould, & Page (2015) proposed an allocation and routing problem for midwives to make a satisfying relationship between mothers and midwives and reduce emerging problems caused by no continuity of care. Clarke and Wright's algorithm was integrated into simulation procedures to examine a trade-off between the travel time and the continuity of care. They indicated that the continuity of care is achievable with small increases in the total traveling time. Wirmitzer, Heckmann, Meyer, & Nickel (2016) proposed five MIP formulations to assess different measures of the continuity of care for a midterm-term home care rostering problem. The continuity improvements resulting from their proposed formulations were compared to a manual solution through experiments on real-world instances. The weekly version of the HHCRSP was addressed through a set partitioning heuristic proposed by Grenouilleau, Legrain, Lahrichi, & Rousseau (2019). A large neighborhood search is developed to solve their proposed set partitioning formulation. It has been shown that this formulation is more promising than the node-based formulations (Bai, Xue, Chen, & Roberts, 2015). The first benefit of this formulation is its capability to handle nonlinear cost functions and nonlinear time-related constraints by hiding them through the generation of feasible route sets. The second advantage is the reduced size of the search space compared with commonly used node-based formulations. However, increasing the number of nodes in this formulation may cause the size of the feasible routes to grow exponentially, and therefore computing the optimal cost of all feasible routes is NP-hard (Xue, Bai, Qu, & Aickelin, 2021).

In recent years, many studies have started to consider problems with uncertainty in the context of the VRPTW (Ghasemkhani, Tavakkoli-Moghaddam, Rahimi, Shahnejat-Bushehri, & Tavakkoli-Moghaddam, 2021), giving rise to robust versions of the VRPTW. Wu, Hifi, & Bederina (2017) developed a novel robust factor for the VRP, in which uncertainty was considered in the travel time. This study aimed to find a robust solution that shows better performance in the majority of scenarios. They analyzed the presented robust criterion on perfectly robust and pseudo robust facts. To provide the former (i.e., perfectly robust conclusion), an exact method was applied for small-sized instances, while for the latter conclusion, an efficient meta-heuristic approach was proposed for solving large-sized problems. The results indicate that the developed criterion in the majority of cases can provide robust solutions. The robust VRP with hard time windows and uncertainty in the travel time was proposed by Braaten, Gjønnnes, Hvattum, & Tirado (2017). To solve the considered NP-hard problem in a reasonable computational time, a heuristic approach based on an adaptive large neighborhood search was presented. This method has been tested on instances with different uncertainty levels, and the results have shown how a high-quality solution could be achieved within short computational times. Lu & Gzara (2019) studied the VRPTW under demand uncertainty. The model ensured the feasibility of each route when demand values on the route changed in the predefined set. They developed a branch-and-price-

and-cut algorithm for solving the majority of problems optimally in a reasonable computational time. Simulation results showed that the robust model provided superior protection against demand uncertainty. Considering the travel time and demand uncertainty and based on uncertainty sets, which are route-dependent, [Hu, Lu, Liu, & Zhang \(2018\)](#) developed a robust optimization model for the VRP with hard time windows. To cope with large-sized problems, they proposed a two-stage methodology based on adaptive VNS. Lastly, they showed the applicability of the presented solution algorithm using benchmark instances. However, in the context of the HHCRSP, there is only one paper that considered robust optimization for this problem by assuming uncertain travel and service times. [Shi et al. \(2019\)](#) proposed a robust model for the HHCRSP by considering uncertain service and travel times. They adapted Gurobi solver, SA, TS, and VNS to solve their suggested model. Then, the strength of considering uncertainties was highlighted by analyzing the Monte Carlo simulation results. Recently, [Fathollahi-Fard, Ahmadi, & Karimi \(2020\)](#) presented a multi-objective mathematical model for an HHCRSP under uncertainty. The proposed model tends to minimize the total cost and total emission generated by the vehicles. Furthermore, a robust optimization model was employed to deal with uncertainty in travel and service times. Finally, different *meta*-heuristic algorithms, including the Keshtel algorithm, NSGA-II, and MOPSO algorithms were used to solve their proposed model. The results indicated the efficiency of the Keshtel algorithm for solving the considered HHCRSP.

The dynamic nature of the problem is another challenge for the HHCRSP. A static version of the problem is that all patients' needs are already known. In the real world, though, patient demand arrives dynamically over the planning horizon ([Demirbilek, Branke, & Strauss, 2019; 2021](#)). This dynamic nature of the problem makes the HHCRSP much more complicated. [Bennett & Erera \(2011\)](#) developed a capacity-based insertion heuristic in a rolling horizon myopic planning approach to solve their proposed dynamic periodic HHCRSP with visit time consistency constraints. [Demirbilek et al. \(2019\)](#) used a scenario-based approach for the dynamic HHCRSP to maximize the average number of daily visits for a single nurse by considering service continuity. [Demirbilek et al. \(2021\)](#) adopted a similar approach by proposing a new heuristic based on generating different scenarios that handle multiple nurses with different skill levels. [Table 1](#) summarizes the main characteristics and assumptions in the reviewed papers.

By reviewing the papers addressed the HHCRSP, it can be concluded that this paper is the first study that considers the dependency between precedence synchronized services and the appropriate level of the continuity of care by assigning distinct caregivers to one patient up to several times in a one-day planning horizon. Moreover, proposing a model that can deploy one caregiver more than once a day is another real-life characteristic that cannot be found in the literature. Furthermore, the reviewed papers reveal that a few studies considered uncertain parameters. However, they have not explored the joint impact of mentioned real-world constraints and robust optimization on the HHCRSP. Also, proposing efficient algorithms for solving and analyzing large-sized robust instances is another challenge addressed in this paper. Moreover, the current literature typically concentrates on the static version that the number of patients and their demands are already known. Thus, to tackle the stated gaps and make the problem more practical, a novel robust optimization model with uncertain service and travel times for the HHCRSP is presented. Subsequently, the three *meta*-heuristic algorithms and Monte Carlo simulation are adapted to analyze the deterministic and robust models. Finally, the dynamic version of this problem is proposed, and two dispatching policies are used to address this problem.

3. Problem description

In this section, a brief description of the proposed problem with the required assumptions is defined. The planning horizon considered for

this problem is one day, which is divided into four periods. According to this division, in each period, each patient can request ordinary services that need only one caregiver or precedence and simultaneous synchronized services that need more than one caregiver. Note that each patient can be visited in their preferred time windows at most twice in each period according to their need for ordinary or synchronized tasks. This approach can provide this possibility for patients with disabilities, who need some services with short durations to be visited up to eight times. Furthermore, this approach can be advantageous for patients who need the continuity of care for their requested services by assigning a specific set of caregivers who are actively communicating with each other and sharing patients' health and care information. It is not possible to provide full continuity of care in all cases by assigning one caregiver to a patient in two non-consecutive periods because of the caregiver's limited working time, which does not cover all the 24 h of a day.

In this paper, two types of caregivers are considered. The first category is employed for performing primary tasks and collecting biological samples from patients. Wound dressing, pain management, medication administration, ostomy care, intravenous therapy, and other health assistance are all examples of primary tasks provided by the first type of caregivers. These caregivers start their tours at the depot, and after delivery of the samples at the laboratory, they finish their tours at the depot in a specific period. Since the laboratory operates independently of the HHC company, all nurses must return to the main depot, which is the HHC company. As stated in the introduction section, a collected biological sample should be transferred to the laboratory before it loses its appropriate condition for the required tests. Therefore, the first category of caregivers is involved in pickup and delivery services. In this case, pick-up nodes are patients' homes, and the delivery node is the laboratory. The time window for visiting the laboratory for each caregiver is determined based on the minimum available time for transferring collected biological samples in the corresponding route. Based on this description, the pickup and delivery problem can be easily converted to the VRPTW. The next category is responsible for synchronized services and, they start and finish their tours at the initial depot without visiting the laboratory. Note that for collecting necessary drugs and medical instruments and also being acquainted with patients' status, the required caregivers for each period need to be at the depot before the starting time of that period. The qualifications of the caregivers for visiting each patient are determined according to the disjunctive services requested by that patient. Therefore, one caregiver with a low qualification level has this eligibility to visit one patient with disabilities, who usually needs a high qualified caregiver in those periods that patient requires only some simple services. [Fig. 1](#) shows an example of the proposed structure for this problem.

3.1. Model notations

The notations of the indices, parameters, and decision variables for the proposed mathematical formulation are summarized as follows:

Indices:

k	Index of caregivers
p	Index of disjunctive services
t, r	Index of periods during a day
i, j	Index of all nodes, $i, j \in \{1, \dots, n\} \cup \{d, f\} \cup \{l\}$. Especially, $1, \dots, n$ represent nodes for patients, d and f with the same location stand for the starting and ending nodes, and l denotes the index of the laboratory.

Sets:

N	Set of all the nodes including a depot, patients, and laboratory
N_c	Set of patients
K	Set of all types of caregivers
K_a	Set of caregivers for primary tasks
K_f	Set of caregivers for synchronized services
P	Set of disjunctive services
P_a	Set of services done by primary caregivers
P_{lab}	Set of laboratory services

(continued on next page)

Table 1
Constraints and assumptions related to the HHCRSP in the literature.

Paper	Synchronized services			Pickup and delivery		Multiple deployments of nurses in one day	Continuity of care	Dynamic setting	Uncertainty				
	Simultaneous sync	Precedence sync	Type of request in one day		Biological sample collection				Sample perishability	Service time	Travel time	Demand	Method
			Single	Multiple									
Bredström & Rönnqvist (2008)	✓	✓	✓										
Bennett & Erera (2011)							✓	✓					
Rasmussen et al. (2012)	✓	✓	✓										
Mankowska et al. (2014)	✓	✓	✓										
Haddadene et al. (2016)	✓	✓	✓										
Eveborn et al. (2006)	✓		✓				✓						
Decerle et al. (2018)	✓		✓										
Decerle et al. (2019a)	✓		✓										
Decerle et al. (2019b)	✓		✓										
Liu et al. (2013)					✓								
Shi et al. (2017)					✓						✓	1	
Shi et al. (2018)					✓					✓	✓	2	
Shi et al. (2019)										✓	✓	3	
Shahnejat-Bushehri et al. (2019)	✓	✓	✓		✓	✓							
Demirbilek et al. (2019)							✓	✓					
Fathollahi-Fard et al. (2020)					✓					✓	✓	3	
Cappanera et al. (2020)	✓	✓	✓										
Demirbilek et al. (2021)							✓	✓					
This Study	✓	✓		✓	✓	✓	✓	✓	✓	✓	✓	3	

Uncertainty method: 1) Fuzzy chance-constraint programming 2) Stochastic programming, and 3) Robust optimization.

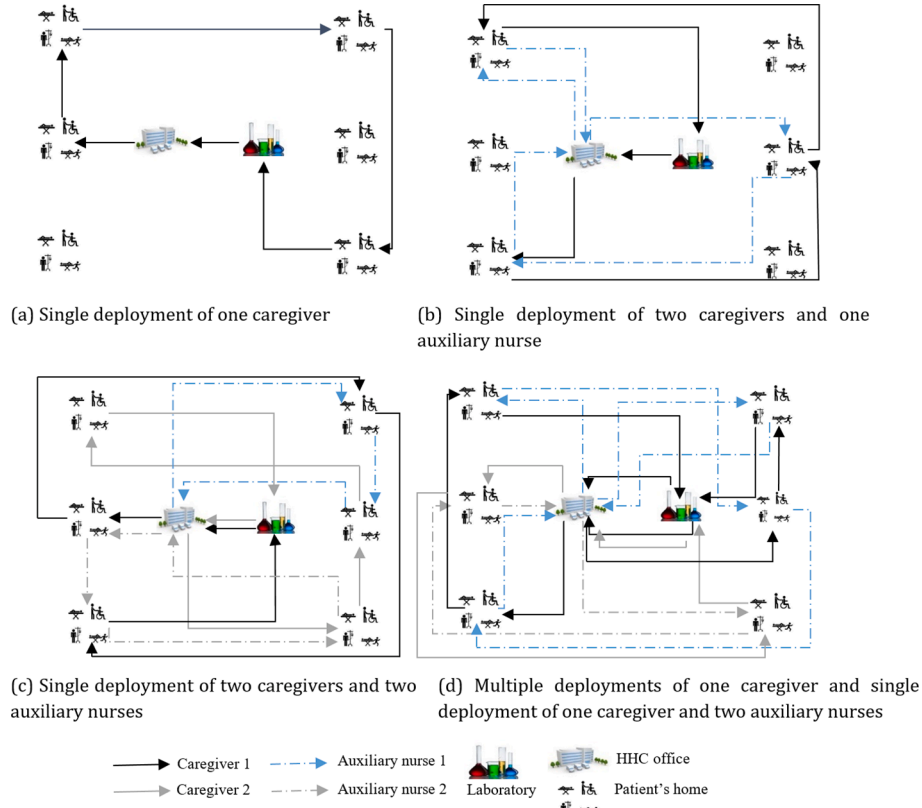


Fig. 1. Example of the proposed structure.

(continued)

P_{sync}	Set of synchronized services
T	Set of time periods during a day
Parameters:	
c_{ij}	Transportation cost to travel from node i to node j based on the distance between nodes
d_{ipt}	1 if patient i needs service p at time period t ; 0, otherwise
tt_{ijt}	Travel time from node i to node j at period t
$[e_{it}, l_{it}]$	Earliest and latest visiting time for patient i who is in demand ($d_{ipt} = 1$) during period t
wt_{it}^k	Required duration for performing services by caregiver type k (primary or auxiliary) for patient i at time period t
pt_{it}	Maximum allowed time for transferring the collected sample from patient i at period t
cs_{tpk}	Cost of performing service p by caregiver k
qu_{pk}	1 if caregiver k has the required qualification for performing service p ; 0, otherwise
$[diff_{it}^1, diff_{it}^2]$	Minimum and maximum gap between synchronized services for the patient i at period t
av_{kt}	1 if the caregiver k is available at two consecutive periods t and $t+1$; 0, otherwise
ft_{kt}	Maximum allowable time for returning caregiver k at period t to be prepared for the next time period
cv_{kt}	Fixed cost of selecting caregiver k for performing services at period t
cc_{itr}	1 if requested services of patient i require continuity of care at periods t and $t+1$; 0, otherwise
M	A very large positive number
Decision variables:	
x_{ijkt}	1 if caregiver k travels from node i to node j at period t ; 0, otherwise
s_{ikt}	Starting time of caregiver k to provide services for patient i at period t
ucv_{kt}	1 if caregiver k is selected for performing services at period t

3.2. Mathematical formulation

In this section, the proposed mathematical model for this study is presented as follows:

$$\text{Min} \sum_{i \in N} \sum_{j \in N} \sum_{k \in K} \sum_{t \in T} c_{ij} x_{ijkt} + \sum_{i \in N} \sum_{j \in N} \sum_{k \in K} \sum_{p \in P} \sum_{t \in T} cs_{tpk} qu_{pk} x_{ijkt} + \sum_{k \in K} \sum_{t \in T} cv_{kt} ucv_{kt} \quad (1)$$

s.t.

$$\sum_{i \in N} \sum_{k \in K_a} x_{ijkt} + \sum_{i \in N} \sum_{k \in K_f} x_{ijkt} \leq 2 \quad \forall j \in N_c, \forall t \in T \quad (2)$$

$$\sum_{j \in N_c} x_{djkt} \leq 1 \quad \forall k \in K, \forall t \in T \quad (3)$$

$$\sum_{i \in N_c} x_{ifkt} \leq 1 \quad \forall k \in K_f, \forall t \in T \quad (4)$$

$$\sum_{i \in N_c} x_{ilkt} \leq 1 \quad \forall k \in K_a, \forall t \in T \quad (5)$$

$$x_{lftk} \leq 1 \quad \forall k \in K_a, \forall t \in T \quad (6)$$

$$\sum_{i \in N} x_{ihkt} = \sum_{j \in N} x_{hjkt} \quad \forall k \in K, \forall h \in N_c, \forall t \in T \quad (7)$$

$$\sum_{i \in N} x_{ilkt} = \sum_{j \in N} x_{ljkt} \quad \forall k \in K_a, \forall t \in T \quad (8)$$

$$\sum_{i \in N} \sum_{k \in K_a} x_{ijkt} \leq \sum_{p \in P_a} d_{ipt} \quad \forall j \in N_c, \forall t \in T \quad (9)$$

$$\sum_{i \in N} \sum_{k \in K_f} x_{ijkt} \leq \sum_{p \in P_{sync}} d_{ipt} \quad \forall j \in N_c, \forall t \in T \quad (10)$$

$$\sum_{p \in P_a} d_{ipt} \leq M \sum_{i \in N} \sum_{k \in K_a} x_{ijkt} \quad \forall j \in N_c, \forall t \in T \quad (11)$$

$$\sum_{p \in P_{\text{sync}}} d_{jpt} \leq M \sum_{i \in N} \sum_{k \in K_f} x_{ijkt} \quad \forall j \in N_c, \forall t \in T \quad (12)$$

$$x_{ijkt} - M(1 - d_{jpt}) \leq q_{upk} \quad \forall i \in N, \forall j \in N_c, \forall k \in K_a, \forall p \in P_a, \forall t \in T \quad (13)$$

$$e_{jt} \sum_{i \in N} x_{ijkt} \leq s_{jkt} \leq l_{jt} \sum_{i \in N} x_{ijkt} \quad \forall j \in N_c, \forall k \in K_a, \forall t \in T \quad (14)$$

$$s_{ikt} + t_{ijt} + w_{it}^k - M(1 - x_{ijkt}) \leq s_{jkt} \quad \forall i, j \in N, \forall k \in K, \forall t \in T \quad (15)$$

$$\begin{aligned} & \text{diff}_{jt}^1 - M(2 - (\sum_{i \in N} x_{ijht} + \sum_{i \in N} x_{ijqt})) \leq \\ & \leq s_{jht} - s_{jqt} \leq \\ & \leq \text{diff}_{jt}^2 + M(2 - (\sum_{i \in N} x_{ijht} + \sum_{i \in N} x_{ijqt})) \end{aligned} \quad (16)$$

$$\forall j \in N_c, \forall h \in K_f, \forall q \in K_a, \forall t \in T$$

$$s_{lkt} - (s_{jkt} + w_{it}^k) - M(2 - d_{jht} - x_{ijkt}) \leq p_{tjt} \quad \forall i \in N, \forall j \in N_c, \forall k \in K_a, \forall h \in P_{lab}, \forall t \in T \quad (17)$$

$$s_{lkt} - f_{tkt} - M(2 - (av_{kt} + \sum_{i \in N} x_{ilkt})) \leq M(1 - x_{djkt(t+1)}) \quad \forall j \in N_c, \forall t \in T, \forall k \in K \quad (18)$$

$$1 - M(2 - (cc_{jtr} + \sum_{i \in N} x_{ijkt})) \leq \sum_{i \in N} x_{ijkr} \quad \forall j \in N_c, \forall t, r \in T, \forall k \in K_a, t < r \quad (19)$$

$$\sum_i \sum_j x_{ijkt} \leq Mucv_{kt} \quad \forall k \in K, \forall t \in T \quad (20)$$

$$x_{ijkt}, uc_{vkt} \in \{0, 1\} \quad \forall i, j \in N, \forall k \in K, \forall p \in P, \forall t \in T \quad (21)$$

$$u_{ikt}, s_{ikt}, id_{ikt} \geq 0 \quad \forall i \in N, \forall k \in K, \forall t \in T \quad (22)$$

The first term in the objective function (1) refers to the minimization of the total traveling cost. The second term minimizes the cost of performing services according to the caregivers' qualifications and experiences. Finally, the last term minimizes the number of caregivers employed for performing services by considering a fixed cost for selecting each caregiver. The first two components are operating costs, while the last term can be considered as a fixed cost. Constraint (2) represents that each patient depending on their need for synchronization services can be visited at most twice at each time period. Constraints (3) and (4) ensure that both types of caregivers start and finish their tour at the determined places, and flow conservation related to the laboratory node is guaranteed by Constraints (5) and (6). The flow conservation and continuity of the routes are respected by Constraints (7) and (8). Constraints (9)-(12) select the patients with demands to be considered for the planning and also exclude the patients with no need at each time period. Constraint (13) determines which caregivers have the required qualification to visit one patient based on his/her types of needs at each time period. Constraint (14) respects patients' time windows. Constraints (15) determines the arrival and starting service time for each patient at each time period and also avoids sub-tours. Constraint (16) ensures that temporal dependencies for patients with demands for synchronized services are considered in the planning.

It should be noted that parameters $[e_{it}, l_{it}]$ are only defined for the primary tasks, and based on the starting time of these tasks, the new time windows are defined for the synchronized services. Transferring the collected biological samples to the laboratory before losing their appropriate condition for required tests is respected by Constraint (17). Constraint (18) restricts the working hours of one caregiver to have enough time to provide essential tools and drugs for the next time period if that caregiver has been assigned to perform services at two

consecutive time periods. The related assignment procedure for patients with a need for the continuity of care for their requested services at different time periods is ensured by Constraint (19). Constraints (20) determines the nurses who are selected for performing services in each period. Constraints (21) and (22) define the nature of the variables and their domain.

3.3. Uncertain travel and service times

In real-world situations, many parameters used in VRPs (e.g., demand, travel time, service time and cost coefficients) because of depending on several factors (e.g., traffic, changes in weather conditions, qualification of the service providers, the patient's condition, the variety and number of services that may be added after the caregiver's visit) are tainted with a high degree of uncertainty. It is meaningful to state when the input data take values different from their nominal values, some constraints may be violated, and therefore, the solution obtained using these parameters may be neither optimal nor feasible (Saeidi-Mobarakeh, Tavakkoli-Moghaddam, Navabakhsh, & Amoozad-Khalili, 2020). Therefore, ignoring uncertainty in the presented mathematical model may lead to a non-optimal solution or even infeasible solution space. To deal with such uncertainty, different approaches have been introduced in the literature. Among them, stochastic programming and robust optimization are two well-known approaches that have widely been applied for incorporating uncertainty into the model. Noteworthy, the former is used when the probability distributions of uncertain parameters are available. However, in real-world situations, there is not enough historical data to estimate the actual distributions of these parameters. As a result, robust optimization approaches that are capable of incorporating decision-making's (DM) risk attitudes are used to cope with the aforementioned drawbacks. In this paper, it is assumed that travel and service times are tainted with uncertainty. Based on Hu et al. (2018), the overall travel time and service time uncertainty sets can be defined by:

$$\vartheta_{tt} = \times_{k \in K} U_{tt}^k \quad (23)$$

$$\vartheta_{wt} = \times_{k \in K} U_{wt}^k \quad (24)$$

$$U_{tt}^k = \left\{ \begin{aligned} & \tilde{t}_{it} \in R^{|N^{k,t}|} \mid \tilde{t}_{ijt} = \hat{t}_{ijt} + \alpha_{ijt} \hat{t}_{ijt}, \\ & \sum_{(i,j) \in A^{k,t}} |\alpha_{ijt}| \leq \Gamma_{tt}^{k,t}, |\alpha_{ijt}| \leq 1, \\ & \Gamma_{tt}^{k,t} \in [0, |A^{k,t}|], \forall (i,j) \in A^{k,t} \end{aligned} \right\} \quad (25)$$

$$U_{wt}^k = \left\{ \begin{aligned} & \tilde{w}_{it} \in R^{|A^{k,t}|} \mid \tilde{w}_{it}^k = w_{it}^k + \beta_{it}^k \hat{w}_{it}^k, \\ & \sum_{i \in N^{k,t}} |\beta_{it}^k| \leq \Gamma_{wt}^{k,t}, |\beta_{it}^k| \leq 1, \\ & \Gamma_{wt}^{k,t} \in [0, |N^{k,t}|], \forall i \in N^{k,t} \end{aligned} \right\} \quad (26)$$

Eqs. (23) and (24) show that ϑ_{tt} and ϑ_{wt} as overall travel time and service time uncertainty are the Cartesian product of uncertain travel and service time sets for each caregiver, respectively. $N^{k,t}$ and $A^{k,t}$ represent the set of patients and arcs on the route of caregiver k in period t , respectively; $\tilde{t}_{ijt}$ and $\tilde{w}_{it}^k$ represent uncertain parameters, which based on Bertsimas & Sim (2004), travel and service times are random variables that their values are in intervals $[\hat{t}_{ijt} - \hat{t}_{ijt}, \hat{t}_{ijt} + \hat{t}_{ijt}]$ and $[\hat{w}_{it}^k - \hat{w}_{it}^k, \hat{w}_{it}^k + \hat{w}_{it}^k]$, respectively; $\tilde{t}_{ijt}$ and $\tilde{w}_{it}^k$ denote the nominal value of uncertain travel time $\tilde{t}_{ijt}$ and service time $\tilde{w}_{it}^k$, respectively; α_{ijt} and β_{it}^k are auxiliary variables; $\hat{t}_{ijt}$ and $\hat{w}_{it}^k$ represent the maximum deviation of uncertain parameters; $\Gamma_{tt}^{k,t}$ and $\Gamma_{wt}^{k,t}$ are "uncertainty budget"

parameters. The above-mentioned parameters adjust the conservatism degree by varying in intervals $[0, |N^{K,t}|]$ and $[0, |A^{K,t}|]$ respectively. Noteworthy, when the value of these parameters is equal to their upper bound, the model acts with the highest level of conservatism, and when they are set to zero, the above constraints are equivalent to that of the nominal problem.

According to the above route-dependent uncertainty sets definition, inspiring from [Agra et al. \(2013\)](#) in adopting resource inequalities, and using adjustable robust optimization presented in [Ben-Tal, El Ghaoui, & Nemirovski \(2009\)](#), the crisp counterpart of the proposed mathematical model can be derived as formulated in Eqs. (27)–(32).

Noteworthy, in adjustable robust optimization, a subset of decision variables are permitted to adjust themselves as the uncertain data change in the uncertainty sets. Accordingly, each variable in the deterministic model is changed to $s_{jkt}(tt, wt)$, where $s_{jkt}(tt, wt)$ is a function of tt and wt . Based on [Agra et al. \(2013\)](#), only travel time vector $tt \in ext(\vartheta_{tt})$ and service time vector $wt \in ext(\vartheta_{wt})$ should be considered in the robust formulation. For more details, the reader is referred to [Agra et al. \(2013\)](#) and [Hu et al. \(2018\)](#). It should be noted that $ext(\vartheta_{tt})$ and $ext(\vartheta_{wt})$ are sets that include all the extreme points of sets ϑ_{tt} and ϑ_{wt} , respectively. Thus, the robust version of the proposed model can be formulated by:

$$\begin{aligned} & \text{Min} \sum_{i \in N} \sum_{j \in N} \sum_{k \in K} \sum_{t \in T} c_{ij} x_{ijkt} \\ & + \sum_{i \in N} \sum_{j \in N} \sum_{k \in K} \sum_{p \in P} \sum_{t \in T} cst_{pk} qu_{pk} x_{ijkt} \\ & + \sum_{k \in K} \sum_{t \in T} cv_{kt} uc_{vkt} \end{aligned} \quad (27)$$

s.t.

Constraints (2) – (13), (19) – (22)

$$\begin{aligned} & e_{jt} \sum_{i \in N} x_{ijkt} \leq s_{jkt}(tt, wt) \leq l_{jt} \sum_{i \in N} x_{ijkt} \\ & \forall j \in N_c, \forall k \in K_a, \forall t \in T, \\ & tt \in ext(\vartheta_{tt}), wt \in ext(\vartheta_{wt}) \end{aligned} \quad (28)$$

$$\begin{aligned} & s_{ikt}(tt, wt) + tt_{ijt} + wt_{it}^k - M(1 - x_{ijkt}) \leq s_{jkt}(tt, wt) \\ & \forall i, j \in N, \forall k \in K, \forall t \in T, \\ & tt \in ext(\vartheta_{tt}), wt \in ext(\vartheta_{wt}) \end{aligned} \quad (29)$$

$$\begin{aligned} & diff_{jt}^1 - M(2 - (\sum_{i \in N} x_{ijht} + \sum_{i \in N} x_{ijqt})) \leq \\ & \leq s_{jht}(tt, wt) - s_{jqt}(tt, wt) \leq \\ & \leq diff_{jt}^2 + M(2 - (\sum_{i \in N} x_{ijht} + \sum_{i \in N} x_{ijqt})) \\ & \forall j \in N_c, \forall h \in K_f, \forall q \in K_a, \forall t \in T, \\ & tt \in ext(\vartheta_{tt}), wt \in ext(\vartheta_{wt}) \end{aligned} \quad (30)$$

$$\begin{aligned} & s_{lkt}(tt, wt) - (s_{jkt}(tt, wt) + wt_{it}^k) - M(2 - d_{jht} - x_{ijkt}) \leq pt_{jt} \\ & \forall i \in N, \forall j \in N_c, \forall k \in K_a, \\ & \forall h \in P_{lab}, \forall t \in T, \\ & tt \in ext(\vartheta_{tt}), wt \in ext(\vartheta_{wt}) \end{aligned} \quad (31)$$

$$\begin{aligned} & s_{lkt}(tt, wt) - ft_{kt} - M(2 - (av_{kt} + \sum_{i \in N} x_{ilkt})) \leq M(1 - x_{djk(t+1)}) \\ & \forall j \in N_c, \forall t \in T, \forall k \in K, \\ & tt \in ext(\vartheta_{tt}), wt \in ext(\vartheta_{wt}) \end{aligned} \quad (32)$$

Constraints (14)–(18) are replaced by constraints (28)–(32) for every $tt \in ext(\vartheta_{tt})$ and $wt \in ext(\vartheta_{wt})$.

4. Solution methods

4.1. Solution representation

As discussed in the previous sections in this paper, there are two types of staff members. Each type has its specific ending structure, and also, there are different time periods in a one-day planning horizon. Considering these features makes the existing commonly used representation methods non-functional for the proposed problem. Therefore, a new encoding method suggested by [Shi et al. \(2017\)](#), which contains a list table for each feasible solution, is modified and adjusted to the proposed problem. With this representation, each solution can simply be converted into a list table, which each row stands for a route of each caregiver, and in the first column of each route, there is a number that shows which caregiver with certain qualifications has been chosen for performing services in a particular period. Furthermore, as shown in [Fig. 2](#), according to the type of the selected caregivers, each type has its ending structure.

4.2. Initial solution

There are sequential and parallel algorithms used in the procedure of the tour-building algorithms for the VRPTW. The sequential method schedules customers by constructing one route at a time. Conversely, in a parallel algorithm, a number of routes are simultaneously scheduled. Before starting these procedures, the number of routes can be considered as a fixed parameter or assumed as a variable that can be changed freely according to the problem ([Solomon, 1987](#)). In this study, a new algorithm based on the free-routing type of parallel procedures is presented that is capable of generating efficient initial solutions in a very short time. The proposed algorithm is applied in two phases. In the first stage, nurses with responsibility for providing primary services are scheduled. This work schedule is determined without considering timing constraints associated with synchronized services. Once a starting time of performing primary tasks for patients with the need for synchronized services is determined, the new time windows are defined for the second tasks.

Subsequently, a work schedule of the auxiliary nurses with responsibility for providing synchronized services is planned. This approach can bring a lower operational cost through wider solution space. Another advantage of this approach is that highly qualified staff members who are usually employed for specialized services are not engaged for some simple activities such as lifting a heavy person, in which a few seconds is needed for two nurses to be together. [Decerle, Grunder, El Hassani, & Barakat \(2016\)](#) introduced this approach first. Since these two stages have the same steps, the proposed method is presented in [Algorithm 1](#) regardless of the type of caregivers.

Consider a given set of patients ($i, j = \{1, \dots, n\}$) that their requested services can be started within a time window $[e_{it}, l_{it}]$. When patient i with service time wt_{it}^k is selected to be allocated to one possible caregiver k , the necessary condition for determining a set of nodes that can be visited after patient i is defined as follows:

If $e_i + wt_{it}^k + tt_j \leq l_j$, then patient j can be added to set $f^{i,t}$ that stands for a set of patients, who can be the next possible alternative in the same route to be visited after patient i . In the next stage, for constructing routes, a criterion should be considered to determine in what order patients should be selected to be inserted into the routes. Since most HHC activities should be implemented at a specified time, HHC companies are faced with tight time windows for their patients. Therefore, set idx^t is defined for each time period, in which patients are sorted based on their latest allowable visiting time.

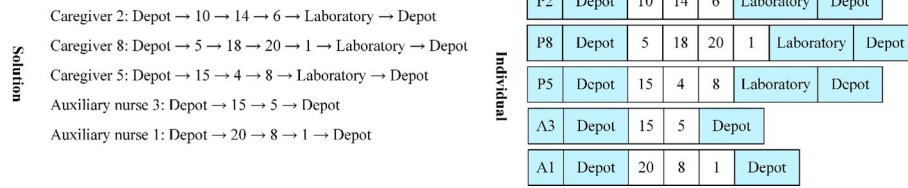


Fig. 2. Example of the solution representation in a particular time period.

Algorithm 1: Method for generating an initial solution

```

1: for  $t = 1$  to 4 do
2:    $sel \leftarrow []$ ,  $tot^t \leftarrow []$ ,  $mi \leftarrow []$ ,  $N \leftarrow$  Number of active patients,  $ii \leftarrow 1$ ,  $r \leftarrow 1$ ,  $route^t\{r\} \leftarrow []$ 
3:   for  $i = 1$  to  $N$  do
4:      $f^{i,t} \leftarrow []$ 
5:     for  $j = 1$  to  $N$  do
6:       if  $e_{it} + w_{it}^k + u_{ijt} \leq l_{jt}$  and  $i \neq j$ 
7:          $f^{i,t} \leftarrow f^{i,t} \cup j$ 
8:       end if
9:     end for
10:   end for
11:   Sort patients based on their latest allowable visiting time and add them to set  $idx^t$ 
12:   Generate predefined routes according to the constraints related to continuity of care and add their nodes to set  $tot^t$ .
13:   while  $length(tot^t) < N$ 
14:      $q \leftarrow 0$ 
15:     if  $\sim isempty(mi)$  ( $mi$  is an empty set)
16:       for  $h = 1$  to  $r$ 
17:         if  $mi(1)$  cannot be found in set  $f^{route^t\{h\}(end),t}$  (the last node of the route  $h$ )
18:            $q \leftarrow q$ 
19:         else
20:            $q \leftarrow q + 1$ 
21:         end if
22:       end for
23:       if  $q$  is equal to 0
24:          $r \leftarrow r + 1$ 
25:          $route^t\{r\} \leftarrow route^t\{r\} \cup mi(1)$ 
26:          $tot^t \leftarrow tot^t \cup mi(1)$ 
27:          $mi(1) \leftarrow []$ 
28:       end if
29:     else
30:       if  $idx^t(ii)$  cannot be found in set  $tot^t$ 
31:         for  $h = 1$  to  $r$ 
32:           if  $idx^t(ii)$  cannot be found in set  $f^{route^t\{h\}(end),t}$ 
33:              $q \leftarrow q$ 
34:           Else
35:              $q \leftarrow q + 1$ 
36:           end if
37:         end for
38:         if  $q$  is equal to 0
39:            $r \leftarrow r + 1$ 
40:            $route^t\{r\} \leftarrow route^t\{r\} \cup idx(ii)$ 
41:            $tot^t \leftarrow tot^t \cup idx(ii)$ 
42:         end if
43:       end if
44:     end if
45:   for  $p = r$  to 1
46:      $sel \leftarrow []$ ,  $K \leftarrow []$ 
47:     Determine the elements of set  $sel$  according to set  $f^{route^t\{p\}(end),t}$ 
48:     Select one node randomly from set  $sel$  and add it to  $K$ 
49:      $tot^t \leftarrow tot^t \cup K$ 
50:      $route^t\{p\} \leftarrow route^t\{p\} \cup K$ 
51:   end for
52:   if  $idx^t(ii)$  cannot be found in  $tot^t$ 
53:      $mi \leftarrow mi \cup idx(ii)$ 
54:   end if
55:    $ii = ii + 1$ 
56: end while
57: Remove the routes with the least number of patients if it is allowed
58: end for

```

With this approach, the patients with lower latest visiting time have a higher possibility of being selected at earlier stages for the scheduling procedure. In the next step, the first route is constructed with the first element of set idx^t , and the selected node is added to a set tot^t . Set sel is temporarily defined, whose elements are determined based on those elements of set $f^{i,t}$ that have not been selected yet. Note that if these elements are added to the corresponding route, the constraints related to patient time windows and the qualification of the caregivers should be respected. The next node is selected randomly from set sel to be added in the constructed route in the previous stage and set tot^t is updated with a new element. Note that primary elements of set sel have a higher probability of being chosen as the next visiting node.

After these steps, the next element of set idx^t should be checked whether it has been chosen in the previous stage. If it has not been chosen yet, this node is added to set mi . In the next iteration, the first element of set mi is checked if it is available in set $f^{i,t}$ for the last node of the constructed routes. If it is not in the stated set, a new route is constructed, and this node is added to the set tot^t . Then, the next nodes are added to the constructed routes based on the last element of each available route. However, if the first element of set mi is available in set $f^{i,t}$, the algorithm is continued based on the stated steps without constructing routes. This process continues until the first element of set mi would be absent in set $f^{i,t}$ for the last element of the constructed routes, and then a new route is constructed. Note that the newly constructed route is assigned at random to a caregiver with a qualification that can satisfy the first patient of that route. This caregiver can also have a higher qualification than it is needed for the stated patient. These steps should be implemented iteratively until all the visiting nodes are added to set tot^t . After finding a feasible solution, the routes with the least number of patients are checked if they can be removed from a solution by inserting them into other routes. According to the constraints related to the continuity of care, in periods $t = \{2, 3, 4\}$, we may start the algorithm with some predefined routes with specified caregivers, which contain the patients who have been visited in the previous time periods. The necessary continuity of care for their current requested services is also tried to be satisfied.

4.3. Solution evaluation and objective function value calculation

Assume that the best-case scenario, in which all uncertain travel and service times are at their nominal values chosen for generating solutions. Although the feasibility of the generated solutions for the best-case scenario is guaranteed, many other scenarios generated solutions are infeasible, and timing constraints related to the time windows of the patients are violated. To diminish such scenarios, a function that returns the latest caregiver arrival time for each patient node should be defined. The intuition behind this notion is that if the proposed meta-heuristic algorithms cannot find a solution that satisfies all the timing constraints according to the given uncertainty sets, the solutions with the least number of violations should be returned (Toklu, 2014).

To calculate the latest caregiver arrival time, consider a given solution $route^t$ that has n routes and $route^t = \{r_1, r_2, \dots, r_n\}$. Let r_k and $|r_k|$ represent the route served by caregiver k and the total number of nodes

on route r_k , respectively. Therefore, each route r_k can be considered as an ordered sequence of patients $r_k = \{p_1^k, p_2^k, \dots, p_{|r_k|}^k\}$. As discussed in Section 3.3, the number of nodes and arcs on a route r_k that has uncertain travel and service times are controlled by $\Gamma_{tt}^{k,t}$ and $\Gamma_{wt}^{k,t}$. Therefore, the latest possible caregiver arrival time can be calculated as a recursive function. This formulation was originally proposed by Shi et al. (2019).

$$\Phi(i, \Gamma_{tt}^{k,t}, \Gamma_{wt}^{k,t}) = \begin{cases} 0 & \text{if } i = d \\ \max \left(e_{it}, \Phi(k, i-1, 0) + \Phi'(k, i-1, 0) + \right. \\ \quad \left. w_{i-1,t}^k + \bar{t}_{i-1,it} \right) & \text{if } i \in N - \{d\}, \Gamma_{tt}^{k,t}, \Gamma_{wt}^{k,t} = 0 \\ \max \left(e_{it}, \Phi(k, i-1, \Gamma_{tt}^{k,t} - 1) + \right. \\ \quad \Phi'(k, i-1, 0) + w_{i-1,t}^k + \\ \quad \bar{t}_{i-1,it} + \hat{t}_{i-1,it}, \\ \quad \Phi(k, i-1, \Gamma_{tt}^{k,t}) + \\ \quad \Phi'(k, i-1, 0) + \\ \quad \left. + w_{i-1,t}^k + \bar{t}_{i-1,it} \right) & \text{if } i \in N - \{d\}, 1 \leq \Gamma_{tt}^{k,t} \leq |r_k|, \Gamma_{wt}^{k,t} = 0 \\ \max \left(e_{it}, \Phi(k, i-1, 0) + \right. \\ \quad \Phi'(k, i-1, \Gamma_{wt}^{k,t} - 1) + \\ \quad w_{i-1,t}^k + \bar{w}_{i-1,t}^k + \bar{t}_{i-1,it}, \\ \quad \Phi(k, i-1, 0) + \Phi'(k, i-1, \Gamma_{wt}^{k,t}) + \\ \quad \left. w_{i-1,t}^k + \bar{t}_{i-1,it} \right) & \text{if } i \in N - \{d\}, 1 \leq \Gamma_{wt}^{k,t} \leq |r_k|, \Gamma_{tt}^{k,t} = 0 \\ \max \left(e_{it}, \Phi(k, i-1, \Gamma_{tt}^{k,t} - 1) + \right. \\ \quad \Phi'(k, i-1, \Gamma_{wt}^{k,t} - 1) + \\ \quad w_{i-1,t}^k + \bar{w}_{i-1,t}^k + \\ \quad \bar{t}_{i-1,it} + \hat{t}_{i-1,it}, \\ \quad \Phi(k, i-1, \Gamma_{tt}^{k,t}) + \Phi'(k, i-1, \Gamma_{wt}^{k,t}) \\ \quad \left. + w_{i-1,t}^k + \bar{t}_{i-1,it} \right) & \text{if } i \in N - \{d\}, 1 \leq \Gamma_{tt}^{k,t}, \Gamma_{wt}^{k,t} \leq |r_k| \\ -\infty & \text{if } i \in N - \{d\}, |r_k| \leq \Gamma_{wt}^{k,t}, \Gamma_{tt}^{k,t} \end{cases} \quad (33)$$

Incorporating the penalty cost of the time windows violation into the considered objective function, the total objective function can be formulated by:

$$\begin{aligned} & \min \sum_{i \in N} \sum_{j \in N} \sum_{k \in K} \sum_{t \in T} c_{ij} x_{ijkt} + \\ & \sum_{i \in N} \sum_{j \in N} \sum_{k \in K} \sum_{p \in P} \sum_{t \in T} c_{st_{pk}} q u_{pk} x_{ijkt} + \\ & \sum_{k \in K} \sum_{t \in T} c v_{kt} u c v_{kt} + \\ & \sum_{k \in K} \sum_{t \in T} \sum_{i \in N}^{r_k} \pi_i \text{Max}(0, \Phi(i, \Gamma_{tt}^{k,t}, \Gamma_{wt}^{k,t}) - l_{it}) \end{aligned} \quad (34)$$

In the above formulation, the first three terms are the same as discussed in the problem description. The last term π_i indicates the penalty coefficient for a time window violation, which is added to calculate the time window violation cost.

4.4. Simulated annealing

Simulated annealing (SA) is a probabilistic method that is used for combinatorial optimization problems. Kirkpatrick, Gelatt, & Vecchi (1983) were among the first researchers to introduce SA for solving these types of optimization problems. In recent years, this method was widely adopted as an efficient approach to approximate global optimization of problems with large search spaces. As mentioned in Section 2, in the context of the HHCRSP, Shi et al. (2018) chose SA among the other four meta-heuristics for solving their proposed stochastic programming model. Also, Fathollahi-Fard, et al. (2018) studied the modification and hybridization of SA and salp swarm algorithm (SSA). Ferreira & de Queiroz (2018) proposed two SA-based heuristics for solving the capacitated location-routing problem. Baniamerian, Bashiri, & Tavakkoli-Moghaddam (2019) used SA as one of their solution approaches to solve a profitable heterogeneous VRP with cross-docking. Wei, Zhang, Zhang, & Leung (2018) presented the SA algorithm with a cooling and rising temperature mechanism for solving the capacitated VRP with two-dimensional loading constraints. Overall, SA can be considered as a robust and efficient meta-heuristic algorithm that can be adopted and modified for solving routing and scheduling problems. The detailed steps for implementation of the proposed SA are discussed below.

4.4.1. Neighborhood structure

As discussed in the previous sections, HHC services requested by the patients should be performed within restricted time windows. In the preserved order of the visits, it is almost impossible for the classical improvement operators (e.g., 2-opt, 2-opt*, relocate move, exchange operator, and other commonly used operators), which have only random nature in their procedure, to make significant improvements in a short time. It should be noted that our proposed problem has four time periods and constraints related to the qualification of the caregivers, which constructing neighbor solutions can be completely time-consuming if an efficient operator is not adopted. To avoid this problem, a new modified version of the insertion operator is presented, which generates guaranteed feasible neighbor solutions in a short time.

For each time period, set rnd^t is randomly generated, which contains the patients with needs for HHC services. Each element of set rnd^t is checked for two possible scenarios to be inserted in other positions based on their defined sets $f^{i,t}$ in Section 4.2 and removed from their current place. In the first scenario, each constructed route is checked if it contains node pa , which can be found in set $f^{i,t}$ for the currently selected node *currentnode* that has been chosen for the insertion procedure from set rnd^t . If timing constraints and other restrictions related to the qualification of the assigned caregiver to that route are not perturbed, node *currentnode* is simply inserted before node pa and removed from its originated route. If this procedure cannot be implemented, in the second scenario, the previous node of node pa is removed and substituted with *currentnode*, and the removed node is checked if it can be inserted in another route based on the procedure that explained in scenario one. If all the constraints are met, the neighbor solution has been successfully

generated. The mechanism of the modified insertion operator can be depicted in Fig. 3. Algorithm 2 describes the detailed steps for implementing this operator.

Algorithm 2: Modified insertion operator

```

1: for  $t = 1$  to 4 do
2:   Select a set of patients randomly and add them to set  $rnd^t$ 
3:    $initialsol \leftarrow route^t$  (the solution that has been selected for modification is firstly added to  $initialsol$ )
4:   for  $u = 1$  to  $length(rnd^t)$  do
5:      $currentnode \leftarrow rnd^t(u)$ ,  $check \leftarrow 0$ 
6:      $rn \leftarrow$  the route number, where node  $currentnode$  is located;  $pn \leftarrow$  the position of node  $currentnode$  in route  $rn$ 
7:     Remove  $currentnode$  from its current position ( $route^t\{rn\}(pn) \leftarrow []$ )
8:     for  $s = 2$  to  $length(f^{currentnode,t})$  do
9:       for  $j = 1$  to  $length(route^t)$  (the number of routes for solution  $route^t$ ) do
10:        if  $check == 0$  and  $j \sim rn$  and  $f^{currentnode,t}(s) \in route^t\{j\}$ 
11:           $pa \leftarrow f^{currentnode,t}(s)$ 
12:          if  $currentnode$  can be inserted in  $route^t\{j\}$ 
13:            Add  $currentnode$  to  $route^t\{j\}$ 
14:             $check \leftarrow 1$ 
15:          elseif  $currentnode$  can be inserted in  $route^t\{j\}$  after removing the previous node of  $pa$  in  $route^t\{j\}$  and  $pa$  is not at the first position in  $route^t\{j\}$ 
16:             $subnode \leftarrow$  the corresponding node that should be removed
17:            Add  $currentnode$  to  $route^t\{j\}$  after implementing the stated condition
18:          for  $ss = 2$  to  $length(f^{subnode,t})$  do
19:            for  $jj = 1$  to  $length(route^t)$  do
20:              if  $check == 0$  and  $jj \sim j$  and  $f^{subnode,t}(ss) \in route^t\{jj\}$ 
21:                 $pa \leftarrow f^{subnode,t}(ss)$ 
22:                if  $subnode$  can be inserted in  $route^t\{jj\}$ 
23:                  Add  $subnode$  to  $route^t\{jj\}$ 
24:                   $check \leftarrow 1$ 
25:                end if
26:              end if
27:            end for
28:          end for
29:          if  $check == 0$ 
30:             $route^t \leftarrow initialsol$ 
31:          end if
32:        end if
33:      end if
34:    end for
35:  end for
36: end for
37: end for

```

The algorithm starts by generating an initial solution sol_0 using the discussed procedure in Section 4.2 and continues until a current temperature T_c is lower than the final solution T_f . In the process of evolution, the generation of neighbor solutions will continue until $nlimit_max$ solutions are accepted or $nover_max$ iterations are implemented. To avoid trapping in local optima and enhancing global optimization, non-improved generated solutions are accepted according to the criterion discussed in Algorithm 3.

4.4.2. Pseudo of simulated annealing (SA)

Before presenting the steps of SA in Algorithm 3, some notations are described below.

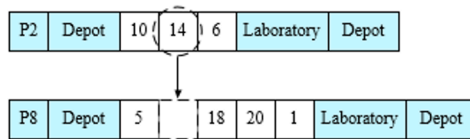


Fig. 3. Example of the mechanism of the modified insertion operator.

T_0 : Initial temperature

T_f : Final temperature

R : Boltzmann's constant value

$nlimit_max$: the maximum number of accepting a new feasible solution at a certain temperature

$nover_max$: the maximum iterations for the local search at a certain temperature

sol_0 : Initial solution

sol^{best} : The best solution that is obtained so far

$sol^{current}$: Current solution

sol^{new} : New solution obtained by a modified insertion operator

$f(sol)$: Objective function value of the solution sol

Algorithm 3: Simulated annealing method

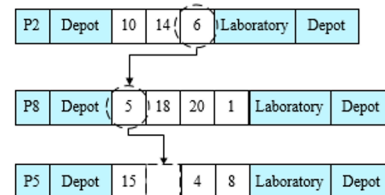
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1: Generate the initial solution  $sol_0$ 
2: Set the parameters for the simulated annealing,  $T_0$ ,  $T_f$ ,  $R$ ,  $nlimit\_max$ ,  $nover\_max$ 
3:  $sol^{best} \leftarrow sol_0$ 
4:  $T_c \leftarrow T_0$ 
5: While  $T_c \geq T_f$  do
6:    $nlimit \leftarrow 0$ ,  $nover \leftarrow 0$ 
7:    $sol^{current} \leftarrow sol^{best}$ 
8:   While  $nlimit \leq nlimit\_max$  and  $nover \leq nover\_max$  do
9:     Generate neighbor solution  $sol^{new}$  from current solution  $sol^{current}$  using the modified insertion heuristic
10:    if  $f(sol^{new}) < f(sol^{current})$ 
11:       $nlimit \leftarrow nlimit + 1$ 
12:       $sol^{current} \leftarrow sol^{new}$ 
13:      if  $f(sol^{current}) < f(sol^{best})$ 
14:         $sol^{best} \leftarrow sol^{current}$ 
15:      end if
16:    else
17:      Randomly generate  $a \in [0, 1]$ 
18:      if  $r < e^{-\frac{(f(sol^{new}) - f(sol^{current}))}{T_c}}$ 
19:         $nlimit \leftarrow nlimit + 1$ 
20:         $sol^{best} \leftarrow sol^{current}$ 
21:      end if
22:    end if
23:     $nover \leftarrow nover + 1$ 
24:  end while
25:   $T_c \leftarrow R * T_c$ 
26: end while

```

4.5. Memetic algorithm

The combination of evolutionary algorithms with local search operators was first introduced by Moscato (1989) named as a memetic algorithm (MA). There are many examples of the successful implementation of this algorithm for solving HHC problems. As can be obtained from the literature section, Decerle et al. (2018) applied the MA for their proposed HHCRSP with soft time windows and synchronization constraints. The efficiency of this algorithm was highlighted by obtaining great results and being adjustable to different types of instances. In another research, Decerle et al. (2019b) considered three conflicting objective functions, including minimization of the total



working time of the nurses, maximization of the quality service, and minimization of the maximal working time difference among caregivers. Similarly, they used the MA to solve this multi-objective HHC problem. Four comparison criteria showed the efficiency of this algorithm in comparison with other existing approaches by obtaining great computational results on benchmark instances. Decerle et al. (2019a) hybridized this algorithm with ant colony optimization (ACO) for their suggested HHC problem. Therefore, the MA as one of the solution approaches in this paper is briefly described in this section.

The MA has one more operator in comparison with the GA, which is a local search operator. It is a population-based method, which starts with the generation of the initial solutions. Based on the represented algorithm in Section 4.2, the initial population is generated. Five main steps, including selection, crossover, mutation, local search, and evaluation should be implemented after generating the initial population. These steps continue to be implemented until the stopping criterion is met. These steps are presented in the remaining part of this section. For breeding a new generation, a selection procedure is performed by selecting individuals from a population. The combination of two techniques proposed by Decerle et al. (2018) is applied to implement this stage. In the first step, a percentage θ_f of the individuals of the next population is chosen based on a fitness proportionate selection procedure. In this step, individuals with lower objective function values have a greater probability of being selected. In the next step, the ranked-based selection procedure is performed to determine the remaining θ_r of population. This procedure is done to avoid trapping into local optimality by reducing selection pressure, which happens in the early stages of running the algorithm when large objective function value differences are common. In this approach, the worst solution is considered as rank 1, and the best solution is assumed as rank N .

Creating new individuals from the population's genetic information is implemented through the crossover operator by reproducing offspring. Since each integer gene is used only once in the solution representation and also the number of routes is dissimilar in each individual, using the traditional crossover operators can lead to generating infeasible offspring which genes are duplicated in one string. Therefore, a new best sub-route crossover operator suggested by Shi et al. (2017), which has no mentioned weaknesses, is adapted to be combined with our individual characteristics. Because this new operator can unavoidably increase the number of routes, after implementing this procedure and obtaining a new offspring, a procedure which was described in Section 4.2 is performed to remove the routes with the least number of patients by checking if they can be removed from a solution by inserting them into other routes. The best sub-route crossover operator is illustrated in Figs. 4 to 7.

Maintaining the genetic diversity of the population is ensured by a

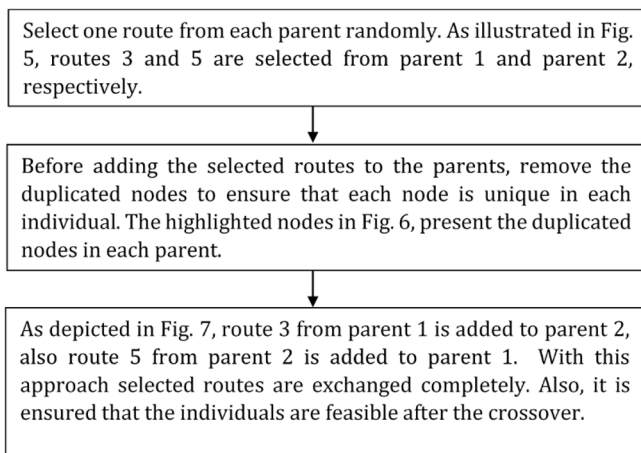


Fig. 4. Example of the main process of the best sub-route crossover.

mutation operator. There are two types of mutation operators that can be undertaken in one route or carried out between two different routes. In this paper, the proposed modified insertion operator in Section 4.4.1 is used as a mutation operator.

Local search operator is implemented iteratively to a part of the population by applying local changes until a time-bound has elapsed. Note that there are two differences between local search and mutation operators (Shi et al., 2017). First, the mutation operator aims to make the population diversified, while the local search aims to improve the solution. Secondly, the local search operator is executed several times, while the mutation operator is implemented only once. In this paper, SA with a limited time-bound is applied to the part of the population as the local search to produce an improvement in the solutions.

The overall process of the proposed MA is shown in Algorithm 4. Initially, a population P is generated by the step given in Algorithm 1. Then, two parents (i.e. p_1 and p_2) are selected from the initial population to be subjected to the crossover operator. Two new offspring chromosomes are produced that are placed into P_{new} ; this continues until the number of chromosomes in P_{new} is equal to *offsize*.

In the next step, a mutation operator is applied to each member of $P \cup P_{new}$ with the probability of α . Similarly, a local search operator is performed on the member of this set with the probability of β . The initial population for the next generation is selected from the best *popsiz*e chromosomes in $P \cup P_{new}$. This process continues until *maxiter* consecutive iterations are implemented without changing in a new best solution.

Algorithm 4: Memetic Algorithm (MA)

```

1:  Set the parameters popsize, offsize,  $\alpha$ ,  $\beta$ 
2:   $P \leftarrow$  popsize initial permutations
3:  While maxiter consecutive iterations have not been implemented without changing in a new best solution do
4:     $P_{new} \leftarrow \emptyset$ 
5:    While  $|P_{new}| \leq \text{offsize}$  do
6:      Randomly select two parents,  $p_1$  and  $p_2$ , from  $P$ 
7:      Produce two children from the parents using the crossover operator
8:      Place new offspring into  $P_{new}$ 
9:    end while
10:    $P' \leftarrow P \cup P_{new}$ 
11:   Apply mutation operator on a member of  $P'$  with probability  $\alpha$ 
12:   Apply local search operator on each member of  $P'$  with probability  $\beta$ 
13:    $P \leftarrow \emptyset$ 
14:   Place the popsize selected solutions from  $P'$  into  $P$ 
15: end while
  
```

5. Experiments

In this section, the design of the test instances to validate the proposed model and algorithms is presented. Firstly, the characteristics of the corresponding instances are described in Section 5.1, then CPLEX solver, SA, GA, and MA are applied to solve the deterministic version of described instances in Section 5.2. After comparing the effectiveness of the proposed algorithms and finding the best approach in Section 5.3, the robust version of the proposed model is considered, and the detailed experiments are presented.

5.1. Experiment description

By reviewing the literature and to the best of our knowledge, the proposed model and assumptions in this research are unique, and therefore, there is no standard benchmark in the literature for this version of the HHCRSP. Solomon's VRTPW benchmark (Solomon, 1987) is adapted to our proposed model, and corresponding experiments are generated.

The instances of this benchmark are classified into six categories, named R1, R2, C1, C2, RC1, and RC2, respectively. These groups are

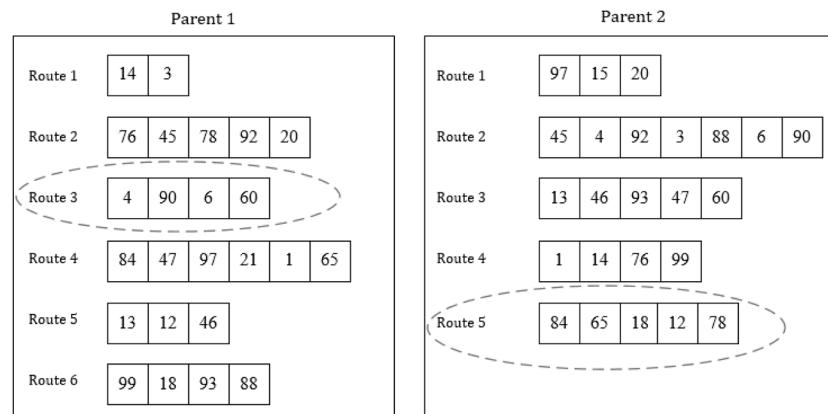


Fig. 5. The first step of the described crossover operator.

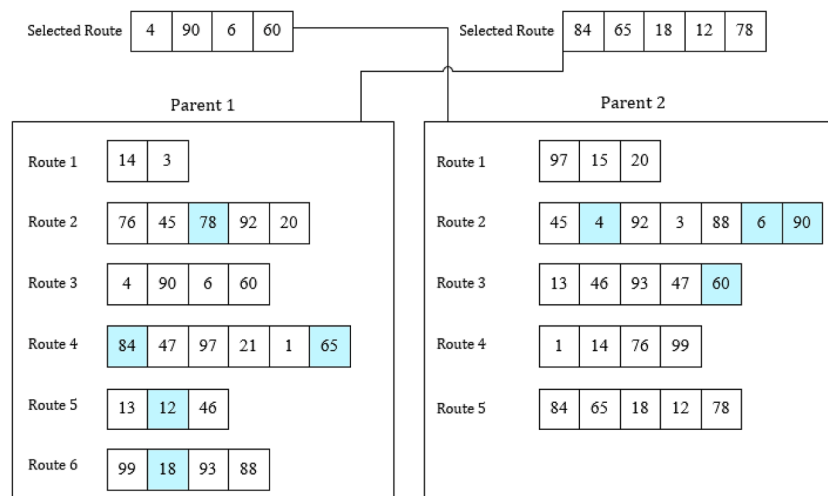


Fig. 6. The second step of the described crossover operator.

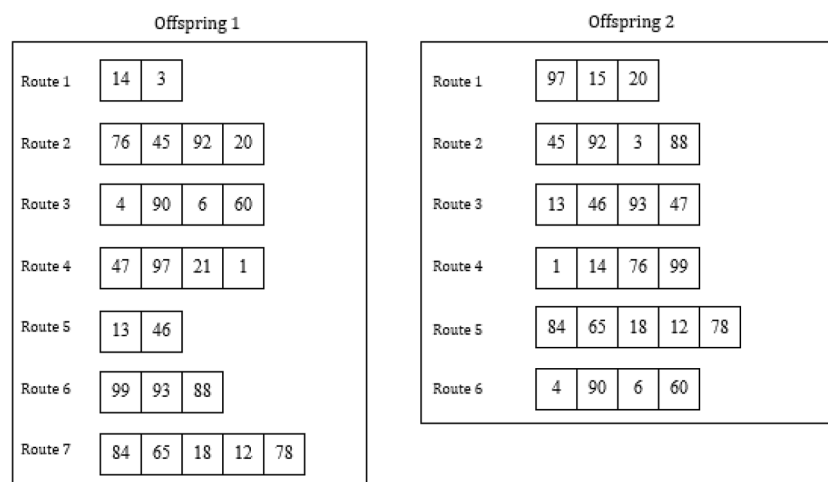


Fig. 7. The third step of the described crossover operation.

separated according to the length of time windows and location distribution. The R instances contain randomly distributed customer locations, while C instances include clustered locations for customers. Also, a mix of clustered and random customer locations is considered in the RC instances. The length of time windows tends to be narrow in the R1, C1, RC1 instances, while wider time windows are considered in the R2, C2,

RC2 instances. The location, demand, time windows, and service time of each customer are defined in each instance of this benchmark. Three sizes of 25, 50, and 100 are considered for the number of customers.

In the proposed problem, similar to Solomon's instances, each patient corresponds to one customer, and every caregiver is assumed to be a vehicle. There are no changes to the customer's location; however, the

following modifications from the original benchmark are considered to adapt it to our proposed model.

- (1) In each instance, one node is added as a location of the lab, whose position is considered to be (35, 45).
- (2) As discussed in Section 2, each patient can request a set of services that need different levels of qualification. Therefore, a set of services is defined, and requested services by each patient in each period are randomly selected from this set. Also, we consider different levels of qualification for each caregiver based on this set.
- (3) The time windows of each patient in period one are similar to Solomon's benchmark; however, in the next periods, the existing time windows from the first period are randomly substituted between patients with demands.

This benchmark is considered to be suitable in the context of the HHCRSP because of the two following reasons (Shi et al., 2019).

- (1) In the HHC services, the location of customers plays a significant role. For example, in some geographical areas, the location of patients tends to be more clustered (C1 and C2 instances). While, in the other areas, the locations are sometimes distributed randomly, so this scenario can correspond to a random type (R1 and R2 instances). In more common situations, the patient's location is a mixture of random and clustered locations, like RC type (RC1 and RC2 instances).
- (2) According to the patients' condition and their preferences, they may prefer a different length of time windows. Therefore, some patients can be serviced in the broader time windows (R1, C1, RC1), while other patients should be serviced in the narrower time windows (R2, C2, RC2).

Consequently, the modified instances of Solomon's benchmark can cover all the necessary features of the HHCRSP. To separate our instances from Solomon's benchmark, instances in this paper are named H-XXX. The main parameters for each algorithm can be obtained from Table 2. Note that the parameters are set empirically. The proposed algorithms are executed on an Intel Core i7 2.00 GHz CPU and 16 GB of RAM by MATLAB 2017. For each instance, we have run each algorithm ten times, then the best solution is reported. Also, GAMS 24.3.3 with CPLEX solver is used to solve small-sized instances.

5.2. Experimental results for the deterministic model

Since there is no similar model in the literature compared to our proposed model, the first group of instances is conducted to show the efficiency of the proposed meta-heuristic algorithms. The first group of experiments is small-sized instances, which can optimally be solved by commercial optimization solvers within an acceptable time. Firstly, GAMS with CPLEX solver is used to solve the problems optimally. Then, SA, GA, and MA are implemented to solve similar instances, and their results are compared with optimal solutions obtained by the CPLEX solver. The results of comparative experiments can be obtained from Table 3, where n , TC , and CT stand for the number of clients, total cost, and computing time, respectively. Also, the specifications column represents the total number of visits, number of synchronous visits, and number of care continuity visits, respectively. The first element of the NV column shows the number of primary nurses, and the second element represents the number of auxiliary nurses. As can be obtained from Table 3, the results obtained by three proposed meta-heuristic algorithms are very close to the exact method solutions, and these algorithms need far less computing time than the CPLEX solver.

Due to the complexity of the problem and the inability of the commercial solvers to solve large-scale problems, the proposed algorithms can be an excellent alternative to find near-optimal solutions within an

Table 2

Parameters used in the algorithms.

Algorithm	Parameters	Values
SA	T_0 : initial temperature	1000
	T_f : final temperature	0.1
	R : Boltzmann's constant value	0.9
	$nlimit_max$: maximum number of accepting a new feasible solution at a certain temperature	100
	$nover_max$: maximum iterations for the local search at a certain temperature	50
MA	$popsiz$: number of chromosomes in the population	200
	$offsize$: number of offspring chromosomes	100
	$maxiter$: maximum number of iterations	400
	θ_f : percentage of the individuals of the next population which is chosen based on the fitness proportionate selection procedure	80%
	θ_r : percentage of the individuals of the next population which is chosen based on the ranked based selection procedure	20%
	α : probability of being selected for a mutation operator	0.2
	β : probability of being selected for a local search operator	0.5
GA	$popsiz$: number of chromosomes in the population	200
	$offsize$: number of offspring chromosomes	120
	$maxiter$: maximum number of iterations	600
	θ_f : percentage of the individuals of the next population which is chosen based on the fitness proportionate selection procedure	80%
	θ_r : percentage of the individuals of the next population which is chosen based on the ranked based selection procedure	20%
	α : probability of being selected for a mutation operator	0.2

acceptable time. Therefore, the proposed methods are applied to 56 instances with a size of 100 patients, and their performances are examined. Tables 4 and 5 report the results of three methods, and the Gap presents the differences among these algorithms. According to the average of total cost and gaps, it can be concluded that the MA is performed better than other algorithms with a bit longer computational time.

5.3. Robust analysis

In this section, the performance of the proposed MA for the robust model is analyzed. Then, we conduct several analyses to investigate the impact of length of time windows, the uncertainty of time windows, the uncertainty of service time, and the location distribution of patients on the achieved robust solutions. To this end, the following indicators are defined to investigate the robustness of the obtained solutions. Let us define V_0 , V_1 , V_2 , and V_3 as the probability of serving all patients at the predefined time window, the probability of not serving at most one client at the predefined time window, the probability of not serving at most two clients at the predefined time window, and the probability of not serving at most three patients at the predefined time window, respectively. It should be noted that the Monte Carlo simulation is implemented to measure the predefined indicators' value. In this procedure, first, we obtain a solution using the described solution method (i.e., the developed MA). Afterward, the average values of the defined simulated indicators obtained from the Monte Carlo simulation are reported as these measures' values.

In this simulation, it is assumed that the service time, as well as the travel time, are trained with uncertainty, and the maximum deviation of these uncertain parameters i.e., travel time and service time from their nominal value, are equal to $0.2 * tt_{ijt}$ and $0.2 * wt_{it}^k$, respectively. Noteworthy, inspiring by the parameter setting conducted by Hu et al. (2018), the coefficients related to the uncertainty budget of $I_{it}^{k,t}$ and $I_{wt}^{k,t}$

Table 3
Experimental results for the small-sized instances by the deterministic model.

ID	n	Specifications			CPLEX			MA			GA			SA		
		n	CT(s)	N.V	TC	CT(s)	N.V	Gap (%)	TC	CT(s)	N.V	Gap (%)	TC	CT(s)	N.V	Gap (%)
HHC-R101	4	(12,2,2)	50.13	(6,2)	308.54	5.02	(6,2)	0.00	308.54	6.27	308.54	0.00	308.54	4.10	(6,2)	0.00
HHC-C101	4	(12,3,3)	44.77	(4,2)	319.16	6.80	(4,2)	0.00	319.16	6.38	319.16	0.00	319.16	4.52	(4,2)	0.00
HHC-RC101	4	(12,4,4)	51.15	(4,3)	290.47	6.16	(4,3)	0.00	290.47	7.31	290.47	0.00	290.47	6.16	(4,3)	0.00
HHC-R201	4	(15,5,2)	46.56	(3,2)	318.88	7.74	(3,2)	0.00	318.88	5.96	318.88	0.00	318.88	5.42	(3,2)	0.00
HHC-C201	4	(15,6,3)	51.36	(3,3)	255.61	5.02	(3,3)	0.00	255.61	7.35	255.61	0.00	255.61	4.45	(3,3)	0.00
HHC-RC201	4	(15,7,4)	40.61	(4,3)	313.97	6.38	(4,3)	0.00	313.97	6.41	313.97	0.00	313.97	5.02	(4,3)	0.00
HHC-R101	8	(20,7,2)	623.24	(6,3)	528.17	52.53	(6,3)	0.00	538.20	43.92	538.20	1.90	543.43	42.98	(7,3)	2.89
HHC-C101	8	(20,8,3)	670.37	(5,4)	572.62	40.75	(5,4)	0.00	583.59	34.33	583.59	1.92	580.73	37.61	(5,4)	1.42
HHC-RC101	8	(20,9,4)	592.86	(4,5)	545.53	53.16	(4,5)	0.00	545.53	49.02	545.53	0.00	558.72	44.32	(4,5)	2.42
HHC-R201	8	(22,9,2)	581.57	(3,3)	499.79	40.53	(3,3)	0.00	512.48	36.77	512.48	2.54	507.14	43.21	(3,3)	1.47
HHC-C201	8	(22,10,3)	581.96	(4,4)	388.06	62.80	(4,4)	0.00	388.06	55.34	388.06	0.00	392.86	51.88	(4,4)	1.24
HHC-RC201	8	(22,11,4)	577.35	(4,4)	529.19	50.35	(4,4)	0.00	541.48	43.22	541.48	2.32	529.19	44.54	(4,4)	0.00

are specified. Accordingly, since we must deal with narrow time windows in R1, RC1, and RC1 instances, the coefficient related to the uncertainty budget of $\Gamma_{it}^{k,t}$ is set to $0.3 \times |N^{k,t}|$.

On the other hand, the time windows in R2, C2, and RC2 instances are wide; therefore, the above-mentioned coefficient related to the uncertain budget of $\Gamma_{wt}^{k,t}$ is set to $0.2 \times |A^{k,t}|$.

5.3.1. MA performance for the robust model

To validate the performance of the solution method developed in this study, we compare its performance in terms of the objective function values with the optimal solutions obtained by solving the robust formulation using the column-and-row generation method addressed in [Agra et al. \(2013\)](#). To this end, the column-and-row generation method was coded in GAMS software, and small-sized instances are solved using CPLEX solver. The results of this comparison are provided in [Table 6](#), which indicates that the developed MA has an acceptable performance in finding optimal solutions for the robust model. As shown in this table, in all instances, the computational time for solving the robust model using the MA is less than the computational time required for solving the presented model using the column-and-row generation method.

5.3.2. Impact of the location distribution of patients

To investigate the effect of location distribution on the achieved solution, the instances in datasets R1, C1, and RC1 are utilized, which have various types of patients' locations. The results are presented in [Table 7](#). Noteworthy, in this analysis, the results that consider the uncertainty of service time and travel time simultaneously are reported, and similar results can be obtained using other patients' locations. The results of this investigation indicate that the achieved deterministic solutions for datasets R1, C1, RC1, especially for R1 and RC1 instances, are very fragile. As reported in [Table 7](#), the average values of V_0 in the R1 and RC1 instances are 0.65 and 0.16 percent, respectively, while this value in the C1 instances is 2.36 percent. Moreover, the average values of V_1 in the R1 and RC1 instances are 2.30 and 0.92 percent, respectively, while this value in the C1 instances is 8.85 percent.

[Table 7](#) also reveals that the non-compliance risk is dramatically decreased by the robust solution. On the other hand, in most cases, the robust model results in an increased total cost, and the decision-maker should investigate the trade-off between the robustness of the obtained solution and total cost.

5.3.3. Impact of the length of time windows

As discussed in the previous section, the main characteristic of the R2, C2, and RC2 instances that makes them different from the R1, C1, and RC1 instances is their wider time windows that result in serving many patients along the same route. The results of deterministic and robust solutions for these instances are described in [Table 8](#). The comparison between the R2, C2, and RC2 instances and the R1, C1, and RC1 instances reveal that the obtained solutions in the former are more robust than the latter. Furthermore, as presented in [Table 8](#), the achieved deterministic solutions are not very fragile (e.g., the values of V_0 for the R2, C2, and RC2 instances are 22.20, 44.62, and 18.81, respectively). Moreover, the values of V_1 for the R2, C2, and RC2 instances are 49.55, 68.96, and 60.14, respectively. Also, the results indicate that the robustness of the obtained solution increases in all cases while the total cost depicted in the last column of [Table 8](#) changes slightly. Also, the values of V_0 obtained for the R2, C2, and RC2 instances, in which both uncertain travel and service times are taken into account, reveal that these solutions are very robust.

Also, as demonstrated in [Table 7](#), in situations where time windows are very tight, the probability of not satisfying all clients is more pervasive in deterministic solutions. Accordingly, obtaining a high degree of robustness will lead to an increase in the objective function. On the other hand, in situations when the time windows are large enough, the acceptable degree of robustness can be obtained without a

Table 4

Experimental results for R1, C1, and RC1 instances with 100 patients by the deterministic model.

Instance	Specifications	Memetic			Genetic				SA			
		TC	CT(s)	N.V	TC	CT(s)	N.V	Gap (%)	TC	CT(s)	N.V	Gap (%)
H-R101	(150,10,40)	8975.03	440.71	(31,3)	8991.29	406.07	(31,3)	0.18	9219.29	392.71	(32,3)	2.72
H-R102	(150,20,30)	9287.55	368.67	(27,5)	9337.95	346.65	(26,6)	0.54	9558.45	311.67	(28,5)	2.92
H-R103	(150,30,20)	8399.16	472.93	(25,6)	8546.73	435.28	(26,6)	1.76	8762.73	429.93	(26,7)	4.33
H-R104	(150,40,10)	8642.04	164.24	(16,9)	8895.89	161.96	(17,9)	2.94	9126.89	121.24	(16,9)	5.61
H-R105	(200,10,90)	9272.46	558.4	(31,5)	9410.96	533.75	(31,5)	1.49	9664.96	510.4	(32,5)	4.23
H-R106	(200,20,80)	11704.32	484.06	(18,8)	11813.24	450.39	(18,8)	0.93	12091.24	414.06	(19,8)	3.31
H-R107	(200,30,70)	11099.52	549.94	(27,8)	11430.74	532.23	(28,8)	2.98	11712.74	506.94	(27,9)	5.52
H-R108	(200,40,60)	6741.50	386.74	(23,10)	7039.20	371.01	(25,10)	4.42	7297.20	321.74	(25,10)	8.24
H-R109	(200,50,50)	7746.90	594.55	(22,11)	7795.32	570.29	(22,11)	0.63	8103.32	549.55	(24,11)	4.60
H-R110	(200,60,40)	9513.48	352.09	(23,11)	9765.84	340.13	(24,11)	2.65	10027.84	287.09	(24,12)	5.41
H-R111	(200,70,30)	9302.64	353.06	(23,9)	9415.38	339.44	(23,9)	1.21	9729.38	290.06	(24,10)	4.59
H-R112	(200,80,20)	10907.36	395.78	(19,10)	11189.26	387.71	(19,10)	2.58	11439.26	332.78	(21,10)	4.88
H-C101	(150,10,40)	4205.79	579.15	(12,4)	4432.59	533.6	(14,4)	5.39	4612.59	526.15	(15,4)	9.67
H-C102	(150,20,30)	4321.62	614.16	(11,4)	4370.90	606.54	(11,4)	1.14	4550.90	569.16	(12,4)	5.31
H-C103	(150,30,20)	4063.23	286.35	(14,6)	4142.48	274.51	(14,6)	1.95	4343.48	226.35	(15,7)	6.90
H-C104	(150,40,10)	4782.65	277.52	(12,5)	5026.43	255.23	(13,5)	5.10	5227.43	227.52	(14,5)	9.30
H-C105	(200,30,70)	4673.10	605.04	(16,5)	4927.54	583.62	(17,5)	5.44	5245.54	560.04	(18,5)	12.25
H-C106	(200,40,60)	7476.82	205.16	(15,6)	7675.24	200.89	(15,6)	2.65	7965.24	142.16	(16,6)	6.53
H-C107	(200,50,50)	4698.10	353.73	(16,5)	5088.72	321.89	(18,5)	8.31	5332.72	304.73	(18,5)	13.51
H-C108	(200,60,40)	4804.50	278.82	(17,7)	4827.84	275.13	(17,7)	0.49	5103.84	238.82	(18,7)	6.23
H-C109	(200,70,30)	6462.82	253.2	(13,10)	6577.88	244.59	(13,10)	1.78	6885.88	189.2	(14,10)	6.55
H-RC101	(150,10,40)	5126.16	266.75	(21,4)	5158.52	248.65	(21,4)	0.63	5358.52	225.75	(22,4)	4.52
H-RC102	(150,20,30)	3598.88	472.87	(17,4)	3815.55	451.9	(18,4)	6.02	4010.55	407.87	(19,4)	11.44
H-RC103	(150,30,20)	3698.94	487.65	(15,5)	3887.94	451.85	(15,5)	5.11	4091.94	420.65	(16,5)	10.62
H-RC104	(150,40,10)	3383.63	413.28	(13,6)	3643.43	400.96	(16,6)	7.68	3877.43	367.28	(16,6)	14.59
H-RC105	(200,30,70)	5454.84	657.06	(25,7)	5773.84	604	(26,7)	5.85	6015.84	587.06	(26,8)	10.28
H-RC106	(200,40,60)	4017.62	582.37	(22,9)	4113.42	574.2	(22,9)	2.38	4353.42	535.37	(23,9)	8.36
H-RC107	(200,50,50)	2902.80	482.65	(20,8)	2944.36	440.97	(20,8)	1.43	3188.36	416.65	(21,8)	9.84
H-RC108	(200,60,40)	3969.02	823.7	(19,10)	4201.54	764.57	(21,10)	5.86	4493.54	774.7	(20,11)	13.22
Total Avg.		6525.26	440.02	(19.41,6.9)	6697.93	417.52	(20.03,6.93)	3.09	6944.48	385.78	(20.72,7.14)	7.43

Table 5

Experimental results for R2, C2, and RC2 instances with 100 patients by the deterministic model.

Instance	Specifications	Memetic			Genetic				SA			
		TC	CT(s)	N.V	TC	CT(s)	N.V	Gap (%)	TC	CT(s)	N.V	Gap (%)
H-R201	(150,10,40)	12084.56	544.36	(13,3)	12164.055	455.36	(13,3)	0.66	12410.055	410.36	(14,3)	2.69
H-R202	(150,20,30)	10208.49	325.98	(11,3)	10310.25	243.98	(11,3)	1.00	10551.75	189.98	(12,3)	3.36
H-R203	(150,30,20)	11468.79	308.82	(11,4)	11480.715	266.82	(11,4)	0.10	11711.715	211.82	(12,4)	2.12
H-R204	(150,40,10)	10274.46	303.99	(5,4)	10509.045	215.99	(6,4)	2.28	10710.05	163.99	(6,4)	4.24
H-R205	(200,20,80)	12695.90	492.61	(12,3)	12841.02	432.61	(12,3)	1.14	13057.02	379.61	(13,3)	2.84
H-R206	(200,30,70)	14882.88	446.44	(10,3)	14944.36	363.44	(10,3)	0.41	15206.36	308.44	(11,3)	2.17
H-R207	(200,40,60)	13797.54	420.25	(8,3)	13823.6	344.25	(8,3)	0.19	14043.60	289.25	(8,3)	1.78
H-R208	(200,50,50)	7657.78	418.32	(9,4)	7853.1	348.32	(10,4)	2.55	8091.10	285.32	(11,4)	5.66
H-R209	(200,60,40)	9057.94	471.55	(8,4)	9243.7	421.55	(9,4)	2.05	9495.70	360.55	(9,4)	4.83
H-R210	(200,70,30)	10439.32	451.15	(7,4)	10446.06	398.15	(7,4)	0.06	10716.80	347.15	(8,4)	2.66
H-R211	(200,80,20)	12036.12	440.76	(10,4)	12354.04	373.76	(11,4)	2.64	12568.04	312.76	(11,4)	4.42
H-C201	(150,10,40)	5449.01	422.82	(7,2)	5511.72	359.82	(7,2)	1.15	5700.72	292.82	(8,2)	4.62
H-C202	(150,20,30)	4375.83	458.09	(8,3)	4397.085	387.24	(8,3)	0.49	4652.09	346.80	(9,3)	6.31
H-C203	(150,30,20)	4299.78	419.08	(9,3)	4347.84	363.08	(9,3)	1.12	4554.84	309.08	(10,3)	5.93
H-C204	(150,40,10)	6327.62	398.52	(8,3)	6496.335	318.52	(9,3)	2.67	6707.84	248.52	(10,3)	6.01
H-C205	(200,30,70)	5474.24	571.85	(11,3)	5530.32	507.85	(11,3)	1.53	5868.32	460.85	(13,3)	7.73
H-C206	(200,40,60)	9631.02	580.19	(10,3)	9907.2	490.19	(11,3)	3.38	10173.20	439.19	(11,3)	6.16
H-C207	(200,50,50)	4964.16	459.84	(8,3)	5104.44	400.84	(10,3)	3.34	5392.44	343.84	(11,3)	9.17
H-C208	(200,60,40)	6192.73	442.04	(10,4)	6362.76	399.04	(11,4)	3.26	6578.76	348.04	(11,4)	6.76
H-RC201	(150,10,40)	5312.04	627.92	(11,2)	5521.38	554.92	(12,2)	3.94	5728.38	503.92	(13,2)	7.84
H-RC202	(150,20,30)	4233.45	562.12	(12,3)	4464.975	516.12	(13,3)	5.47	4680.98	455.12	(14,3)	10.57
H-RC203	(150,30,20)	4232.81	403.05	(7,3)	4285.425	337.05	(7,3)	1.24	4444.43	291.05	(8,3)	5.00
H-RC204	(150,40,10)	4013.30	396.36	(7,3)	4170.66	355.36	(8,3)	3.92	4380.66	299.36	(9,3)	9.15
H-RC205	(200,30,70)	7024.32	550.71	(15,3)	7251.24	491.71	(16,3)	3.23	7509.24	433.71	(16,3)	6.90
H-RC206	(200,40,60)	5175.64	603.39	(16,4)	5526.22	536.39	(18,4)	6.77	5800.22	490.39	(19,4)	12.07
H-RC207	(200,50,50)	3040.84	833.56	(14,4)	3323.96	765.56	(16,4)	9.31	3541.96	723.56	(16,4)	16.48
H-RC208	(200,60,40)	5360.28	780.12	(13,4)	5514.7	729.12	(13,4)	2.88	5842.70	672.12	(14,4)	9.00
Total Avg.		7767.07	486.44	(10.3,3.30)	7914.30	421.37	(10.63,3.30)	2.40	8152.55	367.32	(11.37,3.30)	6.09

significant increase in the considered objective function.

5.3.4. Impact of uncertainty in travel and service times

In this section, different types of the developed model, including deterministic model, defined model under service time uncertainty

(MUSTU), developed model under travel time uncertainty (MUTTU), and developed model under both service time and travel time uncertainty (MUSTTU) are investigated to evaluate the effect of uncertain travel time and service time on the described mathematical model for the instances with 100 patients. The results of these analyses are

Table 6

Comparative results between MA and column-and-row generation algorithm for the robust model.

Instances ID	n	Specification	Column-and-row generation			MA		
			TC	N.V	CT (s)	TC	N.V	CT(s)
HHC-R101	4	(12,2,2)	339.39	(6,2)	41.58	339.39	(6,2)	6.02
HHC-C101	4	(12,3,3)	341.50	(4,2)	36.60	341.50	(4,2)	8.16
HHC-RC101	4	(12,4,4)	334.04	(4,3)	44.92	334.04	(4,3)	7.39
HHC-R201	4	(15,5,2)	363.52	(4,2)	38.45	363.52	(4,2)	10.06
HHC-C201	4	(15,6,3)	276.06	(3,3)	45.15	276.06	(3,3)	7.03
HHC-RC201	4	(15,7,4)	357.93	(4,3)	34.51	357.93	(4,3)	8.93
HHC-R101	8	(20,7,2)	575.71	(6,3)	515.04	575.71	(6,3)	73.54
HHC-C101	8	(20,8,3)	618.43	(5,4)	581.74	618.43	(5,4)	52.98
HHC-RC101	8	(20,9,4)	616.45	(4,5)	516.02	616.45	(4,5)	74.42
HHC-R201	8	(22,9,2)	564.76	(4,3)	498.07	564.76	(4,3)	52.69
HHC-C201	8	(22,10,3)	415.22	(4,4)	478.02	415.22	(4,4)	75.36
HHC-RC201	8	(22,11,4)	555.65	(4,4)	463.26	555.65	(4,4)	65.46

Table 7

Experimental results for R1, C1, and RC1 instances with 100 patients.

Instance	Deterministic model (%)					Robust model (%)					Increase (%)
	TC	V ₀	V ₁	V ₂	V ₃	TC	V ₀	V ₁	V ₂	V ₃	TC
H-R101	5983.35	0.05	0.49	1.15	3.79	6963.95	24.37	38.40	56.13	88.99	16.39
H-R102	6191.70	0.94	4.55	5.53	8.23	7032.06	41.18	74.68	85.18	91.02	13.57
H-R103	5599.44	0.14	0.52	5.71	7.61	6490.75	34.25	62.18	85.40	98.15	15.92
H-R104	5761.36	0.04	0.17	3.13	5.95	6769.12	23.18	76.38	94.26	100.00	17.49
H-R105	4636.23	0.10	1.21	3.58	7.28	4730.70	30.14	48.62	66.21	91.65	2.04
H-R106	5852.16	0.64	1.99	4.24	6.91	6626.06	36.84	85.41	92.00	97.14	13.22
H-R107	5549.76	1.57	3.71	6.19	8.00	5684.64	65.12	75.78	83.97	94.60	2.43
H-R108	3370.75	0.04	0.87	6.24	7.77	3621.35	27.11	43.10	85.41	92.00	7.43
H-R109	3873.45	1.49	5.19	7.00	7.67	4145.95	47.35	60.55	84.29	89.79	7.04
H-R110	4756.74	0.12	1.13	3.37	6.08	5310.60	77.32	91.62	98.00	100.00	11.64
H-R111	4651.32	1.24	2.24	6.51	8.05	4831.98	59.37	73.42	88.21	91.27	3.88
H-R112	5453.68	1.42	5.51	7.60	8.05	5691.52	48.12	64.09	89.00	94.00	4.36
Avg.		0.65	2.30	5.02	7.12		42.86	66.19	84.01	94.05	
H-C101	2803.86	0.45	6.35	6.79	7.75	3187.71	62.35	81.68	92.85	97.63	13.69
H-C102	2881.08	1.12	8.94	10.26	10.46	3572.41	64.86	80.00	87.00	94.83	24.00
H-C103	2708.82	2.34	11.09	11.47	12.11	3919.93	70.60	92.00	96.00	99.80	44.71
H-C104	3188.43	2.50	11.22	11.60	12.70	3772.15	82.44	98.00	100.00	100.00	18.31
H-C105	2336.55	3.52	9.06	9.92	11.21	2684.54	72.23	80.21	83.72	88.83	14.89
H-C106	3738.41	1.92	6.30	6.95	9.60	5589.85	74.06	83.00	92.00	97.40	49.52
H-C107	2349.05	2.72	8.87	10.20	11.57	2707.77	72.04	81.20	88.48	94.23	15.27
H-C108	2402.25	3.49	8.74	10.15	11.72	2740.79	72.60	77.28	86.38	98.97	14.09
H-C109	3231.41	3.17	9.05	10.19	11.82	3657.89	71.81	83.84	87.99	99.48	13.20
Avg.		2.36	8.85	9.72	10.99		71.44	84.13	90.49	96.80	
H-RC101	3417.44	0.05	0.25	1.45	2.17	4156.05	38.89	54.26	72.20	82.61	21.61
H-RC102	2399.25	0.03	0.63	1.25	1.87	3009.01	34.00	43.59	65.00	74.02	25.41
H-RC103	2465.96	0.51	2.15	2.82	3.48	2864.21	56.20	65.13	86.18	92.50	16.15
H-RC104	2255.75	0.08	1.14	2.72	3.07	2431.61	32.42	48.99	79.81	81.82	7.80
H-RC105	2727.42	0.13	0.74	2.22	3.49	3092.52	37.20	58.88	86.88	94.18	13.39
H-RC106	2008.81	0.16	0.62	1.13	1.62	2067.24	40.99	52.38	77.62	89.56	2.91
H-RC107	1451.40	0.12	0.62	2.16	2.94	1667.17	38.66	57.54	72.87	88.85	14.87
H-RC108	1984.51	0.24	1.22	3.35	3.59	2226.84	36.42	50.23	91.98	97.38	12.21
Avg.		0.16	0.92	2.14	2.78		39.35	53.88	79.07	87.62	
Total Avg.		1.05	3.95	5.69	7.12		50.76	68.36	84.66	93.13	

presented in Table 9. The obtained results, which are the average value of the predefined indicators V_s reveal that the deterministic model has the lowest robustness while the developed model under both travel time and service time uncertainty has the maximum robustness. For example, in RC1 instances, the value of V_1 in the defined models is 0.92, 13.02, 18.67, and 53.88, respectively.

We also conduct some comparisons between the robust and deterministic models to investigate the efficiency of the developed robust mathematical model. Accordingly, the value of the defined indicators V_s obtained through the described simulation procedure for the 100-client instances is provided in Figs. 8 to 11. These figures indicate that the obtained values of the defined indicators using the developed robust mathematical model are almost higher than those achieved through the deterministic one.

5.4. Dynamic extension

In this section, the dynamic version of the home health care routing-scheduling problem (DHHCRSP) is presented. Due to the similarity of our considered problem with the ridesharing problem, the same approach developed by Sun, Chen, & Zhang (2020) is implemented to solve the proposed DHHCRSP. In this case, a given planning horizon $[0, T]$ is considered for each time period. Some patient requests are received before time 0, and others are received randomly along the planning horizon. The matching outcome for each patient and caregiver must be determined before a specified notification time in each period, denoted as α_i^t and α_k^t . These times must be less than the latest departure time for a caregiver and the latest visiting time for a patient. In a dynamic situation, the HHC company does not have much information about future demands, and new requests must be planned quickly because each

Table 8

Experimental results for R2, C2, and RC2 instances with 100 patients.

Instance	Deterministic model (%)					Robust model (%)					Increase (%)
	TC	V0	V1	V2	V3	TC	V0	V1	V2	V3	
R201	8056.37	10.76	47.91	74.19	86.51	8320.53	87.78	93.98	99.98	100.00	3.28
R202	6805.66	10.32	23.25	33.31	45.80	6921.96	89.03	92.11	99.00	100.00	1.71
R203	7645.86	18.15	69.77	89.60	94.01	7658.77	88.73	90.72	98.69	99.99	0.17
R204	6849.64	22.63	62.80	91.40	97.76	6995.80	90.52	92.20	100.00	100.00	2.13
R205	6347.95	54.44	76.73	81.63	93.89	6527.12	90.95	94.32	100.00	100.00	2.82
R206	7441.44	16.12	36.28	47.55	59.86	7699.68	93.69	97.91	100.00	100.00	3.47
R207	6898.77	39.18	56.67	70.39	82.85	7356.41	95.75	98.33	100.00	100.00	6.63
R208	3828.89	19.46	60.33	74.08	78.62	4029.07	97.30	99.33	100.00	100.00	5.23
R209	4528.97	11.36	31.97	73.92	86.29	4919.28	93.28	96.37	100.00	100.00	8.62
R210	5219.66	18.21	31.55	35.89	45.37	5341.78	90.71	97.94	99.93	100.00	2.34
R211	6018.06	23.59	47.82	57.95	70.45	6148.34	93.77	97.64	100.00	100.00	2.16
Avg.		22.20	49.55	66.36	76.49		91.96	95.53	99.78	100.00	
C201	3632.67	49.42	66.54	95.33	98.99	3690.87	89.46	97.59	100.00	100.00	1.60
C202	2917.22	47.15	63.62	99.18	100.00	3035.80	96.11	98.03	100.00	100.00	4.06
C203	2866.52	45.75	65.29	98.09	100.00	2924.09	95.67	99.79	100.00	100.00	2.01
C204	4218.41	48.42	70.82	95.20	99.87	4337.73	97.59	99.42	100.00	100.00	2.83
C205	2723.5	42.89	71.59	95.49	98.47	2829.87	94.39	99.68	100.00	100.00	3.91
C206	4791.55	47.06	70.40	94.86	98.08	4914.45	96.99	98.15	100.00	100.00	2.56
C207	2469.73	28.28	71.01	97.05	99.31	2624.14	93.66	97.05	100.00	100.00	6.25
C208	3080.96	47.98	72.40	94.38	99.74	3172.42	95.10	98.95	100.00	100.00	2.97
Avg.		44.62	68.96	96.20	99.31		94.87	98.58	100.00	100.00	
RC201	3541.36	14.58	58.48	89.62	96.62	3718.43	93.03	99.81	100.00	100.00	5.00
RC202	2822.3	18.10	65.14	92.74	97.70	2934.95	94.30	98.43	100.00	100.00	3.99
RC203	2821.87	21.03	54.98	93.19	97.70	2935.17	89.13	97.93	100.00	100.00	4.02
RC204	2675.53	16.38	63.61	88.83	97.34	2809.24	92.84	96.32	100.00	100.00	5.00
RC205	3512.16	17.11	50.15	90.56	96.72	3535.44	90.91	99.35	100.00	100.00	0.66
RC206	2587.82	20.98	62.12	93.32	97.70	2717.21	94.12	95.75	100.00	100.00	5.00
RC207	1520.42	19.17	60.17	88.99	97.68	1597.93	90.34	97.16	100.00	100.00	5.10
RC208	2680.14	23.12	66.44	91.00	97.70	2823.90	94.80	98.47	100.00	100.00	5.36
Avg.		18.81	60.14	91.03	97.40		92.43	97.90	100.00	100.00	
Total Avg.		27.84	58.44	82.51	89.45		92.96	97.14	99.91	100	

Table 9

Results of achieved solutions by deterministic, MUSTU, MUTTU, and MUSTTU models.

Model type	Average (%)	R1	C1	RC1	R2	C2	RC2
Deterministic	Avg. V ₀	0.65	2.36	0.16	22.20	44.62	18.81
	Avg. V ₁	2.30	8.85	0.92	49.55	68.96	60.14
	Avg. V ₂	5.02	9.72	2.14	66.36	96.20	91.03
	Avg. V ₃	7.12	10.99	2.78	76.49	99.31	97.40
MUSTU	Avg. V ₀	8.12	18.08	7.98	57.49	81.22	56.87
	Avg. V ₁	15.00	30.88	13.02	73.54	92.14	88.20
	Avg. V ₂	23.90	34.78	48.65	87.22	97.59	93.14
	Avg. V ₃	40.88	41.43	61.00	97.11	99.71	97.58
MUTTU	Avg. V ₀	14.23	21.73	5.66	78.17	85.78	81.14
	Avg. V ₁	35.55	41.97	18.67	83.64	93.15	90.59
	Avg. V ₂	39.40	47.91	32.80	87.07	97.05	92.00
	Avg. V ₃	51.71	54.61	56.78	96.62	99.67	98.70
MUSTTU	Avg. V ₀	42.86	71.44	39.35	91.96	94.87	92.43
	Avg. V ₁	66.19	84.13	53.88	95.53	98.58	97.90
	Avg. V ₂	84.01	90.49	79.07	99.78	100.00	100.00
	Avg. V ₃	94.05	96.80	87.62	100.00	100.00	100.00

MUSTU: Model under service time uncertainty; MUTTU: Model under travel time uncertainty; MUSTTU: Model under both service time and travel time uncertainty

patient has his/her time constraints and needs.

To deal with the dynamic version of this problem, a given planning horizon is divided into B epochs. The starting time of each epoch b is indicated by t_b , which is considered as a decision point. The ending time is $t_{b+1} = t_b + \tau$, where τ is the length of each epoch. At starting time of each epoch, all available information up to this point is collected, and static problem Π_b is solved. Each static problem Π_b has an allowed computation time δ , and $t_b + \delta$ is defined as a dispatching point. At this point, the whole or part of the solution obtained from solving Π_b at decision point t_b is implemented. The static problem Π_b consists of all the caregivers and patients that have not been dispatched and visited by t_b , and their notification time is after than $t_b + \delta$. In medical services, since it is necessary to take the necessary drugs and equipment according to the patients' needs, once a caregiver is dispatched, the caregiver route is

final, and no new patients can be assigned to this caregiver. However, if a caregiver is not dispatched, new dynamic requests can be inserted into its candidate route when the problem is resolved.

In this paper, greedy and postponement policies, which have been suggested by Sun, Chen, & Zhang (2020), are adopted at each dispatching point. In the first policy, the static problem Π_b is solved and its entire solution is implemented immediately at the corresponding dispatching point. In the second policy, after solving the static problem, two criteria are checked to see if the selected route for one caregiver is still feasible for the next dispatching point, and $\alpha'_k \geq t_b + \delta$ and $\alpha'_k \geq t_b + \delta$ for the caregiver and every patient on this route. By satisfying these two conditions, this caregiver can be dispatched to the next dispatching point. Given this policy, this caregiver is considered again in the next problem Π_{b+1} at the next decision point. Otherwise, the planned route

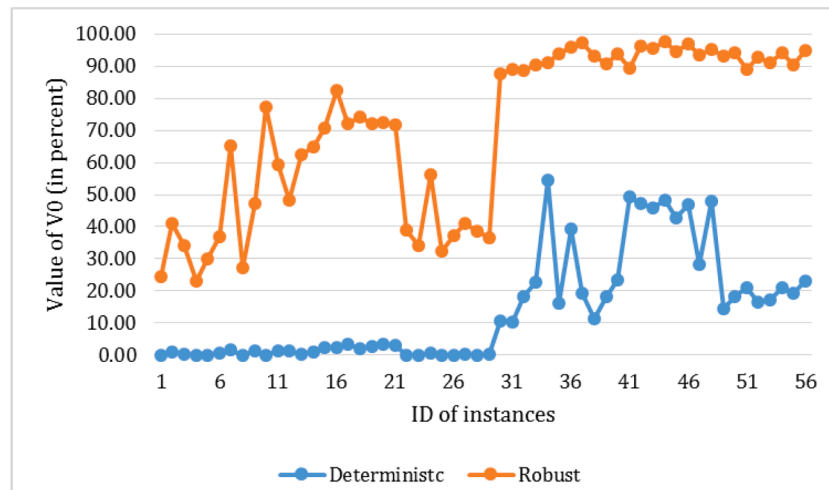


Fig. 8. Value of V_0 obtained by the deterministic and robust models for instances with 100 patients.

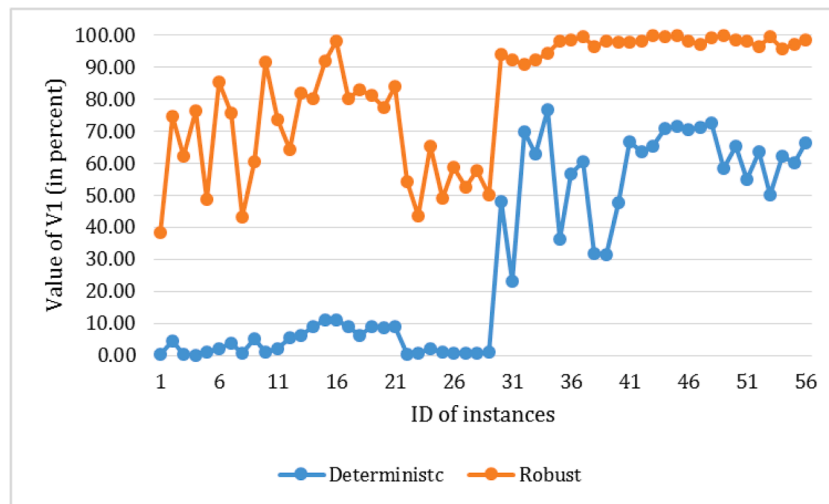


Fig. 9. Value of V_1 obtained by the deterministic and robust models for instances with 100 patients.

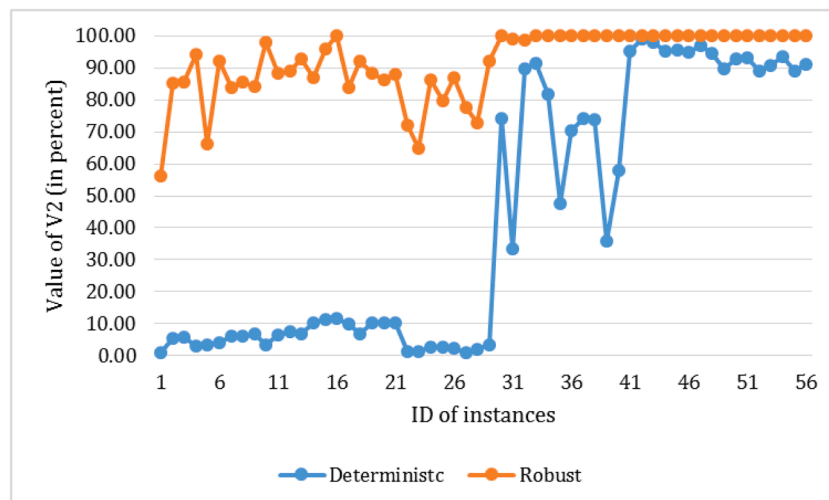


Fig. 10. Value of V_2 obtained by the deterministic and robust models for instances with 100 patients.

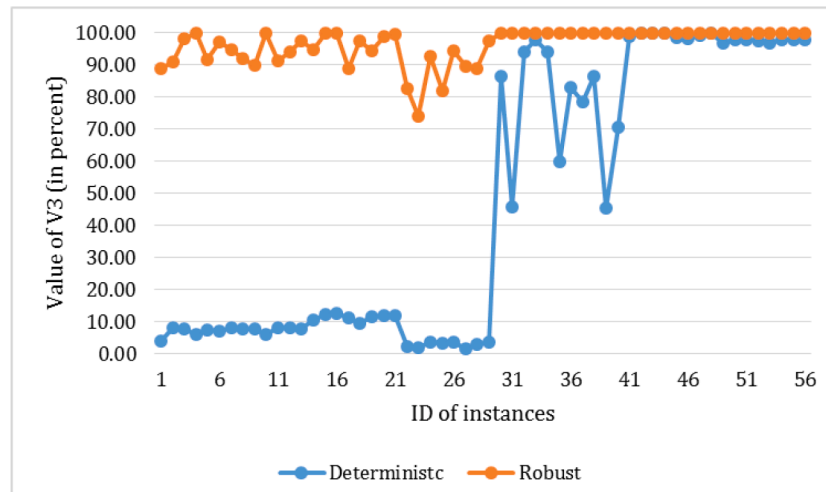


Fig. 11. Value of V_3 obtained by the deterministic and robust models for instances with 100 patients.

for this caregiver is implemented at the current dispatching point. For more details, the reader is referred to Sun, Chen, and Zhang (2020). It should be noted that in both policies, for any caregiver or patient considered in problem Π_b , which is not on any implemented route at time $t_b + \delta$, the caregiver or patient is notified that no matching can be found if the notification time is $\alpha_k^i < t_b + \delta$ or $\alpha_k^i < t_b + \delta$. Otherwise, they can be considered again in problem Π_{b+1} , along with the requests that arrive after t_b and before t_{b+1} and their notification time is after $t_{b+1} + \delta$.

Two approaches can be adopted for solving the static problem Π_b at decision point t_b . In the first approach, the generated route from solving problem Π_{b-1} can be used after some modifications. The second approach is that the problem is solved from scratch without using the generated route in the previous stage. In this study, the generated routes from solving problem Π_{b-1} , which are still valid for the problem Π_b and the corresponding caregivers have not been dispatched at dispatching point $t_{b-1} + \delta$ are imported as an initial solution for problem Π_b . Each new request is also added to the problem Π_b according to the stated conditions. In the next step, the MA is adopted to solve the problem, and the maximum allowed computation time is set to δ .

Based on the above-mentioned description, we provide a comparative analysis to evaluate the performance of the discussed policies. For this purpose, R1, C1, RC1, R2, C2, and RC2 instances are employed, and the performance of the dynamic approach is evaluated under the complete information, postponement policy, and greedy policy. Noteworthy, it is assumed that the requests of 100 patients are known and therefore can be considered together as a static problem discussed in Section 3.

Also, we assume that 20 requests are received over the planning horizon. In the complete information case, it is assumed that all the dynamic and static requests are known before the planning horizon. The other parameters are the same as described in the previous sections. Table 10 shows that the postponement policy has a better performance than the greedy policy. The maximum increase in the total cost is 16.91% for the postponement policy, while this value is 30.29% in the greedy policy. Table 10 also indicates that the increase in R1, C1, and RC1 instances is less than R2, C2, and RC2 instances due to the wider time windows in R2, C2, and RC2 instances. In other words, this time window determines the caregivers' flexibility. Therefore, their schedule becomes more flexible when the time window is wider.

6. Conclusion

In this paper, some real-life assumptions (e.g., the dependency between precedence synchronized services and appropriate level of the continuity of care) were added into the home health care routing-scheduling problem (HHCRSP) to make the problem closer to the real world. Multiple deployments of one HHC caregiver and the possibility of visiting one patient multiple times during different periods of a day were other characteristics that were considered in our proposed mathematical model. To deal with uncertain service and travel times, a robust version of the HHCRSP was presented. Due to the complexity of the problem, three meta-heuristics (i.e., SA, GA, and MA) are developed to solve large-sized problems. The proposed methods showed their efficiency by obtaining near-optimal solutions for small-sized instances, and the commercial solver was able to solve the problems within an acceptable time. However, the MA was more efficient in large-sized problems. Comprehensive analyses were conducted on robust and deterministic models using Monte Carlo simulation, and the results showed that generating solutions without considering uncertainties lead to fragile solutions and often result in infeasible solutions. Finally, the dynamic version of this problem was proposed, and two dispatching policies (including postponement and greedy policies) were used to address this problem. As predicted, the postponement policy outperformed the greedy policy in all instances.

The workload balance, patient preferences, and also uncertainty for patients' demands are interesting characteristics that can make the problem more realistic. Furthermore, developing efficient exact methods to solve the model and comparing them with meta-heuristic algorithms can be another research direction.

Table 10
Comparative results of various strategies.

ID	Complete information	Postponement policy		Greedy policy	
	Average of TC	Average of TC	Increase (%)	Average of TC	Increase (%)
HHC-R1	6,865.52	7579.75	10.40%	8727.33	27.12%
HHC-C1	4,451.22	5204.05	16.91%	5694.76	27.94%
HHC-RC1	3,408.17	3976.31	16.67%	4440.36	30.29%
HHC-R2	7,798.98	8642.57	10.82%	9572.59	22.74%
HHC-C2	4,297.40	4555.26	6.00%	5154.11	19.94%
HHC-RC2	4,297.40	4427.44	3.03%	5040.23	17.29%

CRediT authorship contribution statement

Sina Shahnejat-Bushehri: Conceptualization, Investigation, Writing - original draft. **Reza Tavakkoli-Moghaddam:** Methodology, Supervision, Writing - review & editing. **Mehdi Boronoos:** Project administration, Software, Validation. **Ahmad Ghasemkhani:** Data curation, Formal analysis, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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